

Fundamentals of Solid State Physics

Metals and Insulators

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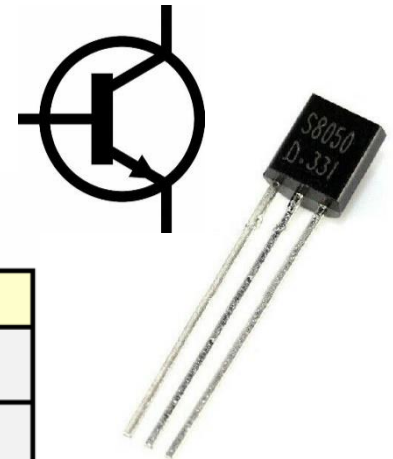
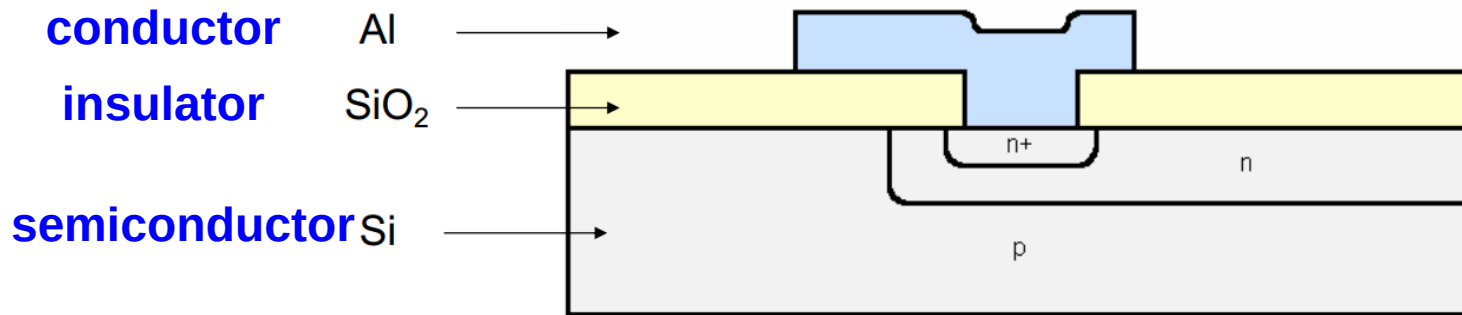
This Class

- Introduction (Week 1)
- Materials and Crystal Structures (Week 2–3)
- **Electronic Properties (Week 4–12)**
 - Free electrons (the Drude and Sommerfeld models)
 - Electrons in a periodic potential (Bloch's Theorem)
 - The near-free electron model, the tight-binding model
 - Electronic band diagram, band gaps, effective mass
 - Metals, insulators, semiconductors
 - Devices: junctions, diodes, transistors
- Thermal Properties (Week 13)
- Optical Properties (Week 14)
- Magnetic Properties (Week 15)

Electronic Properties of Materials

CMOS transistor

- Complementary **Metal-Oxide-Semiconductor**



Metal

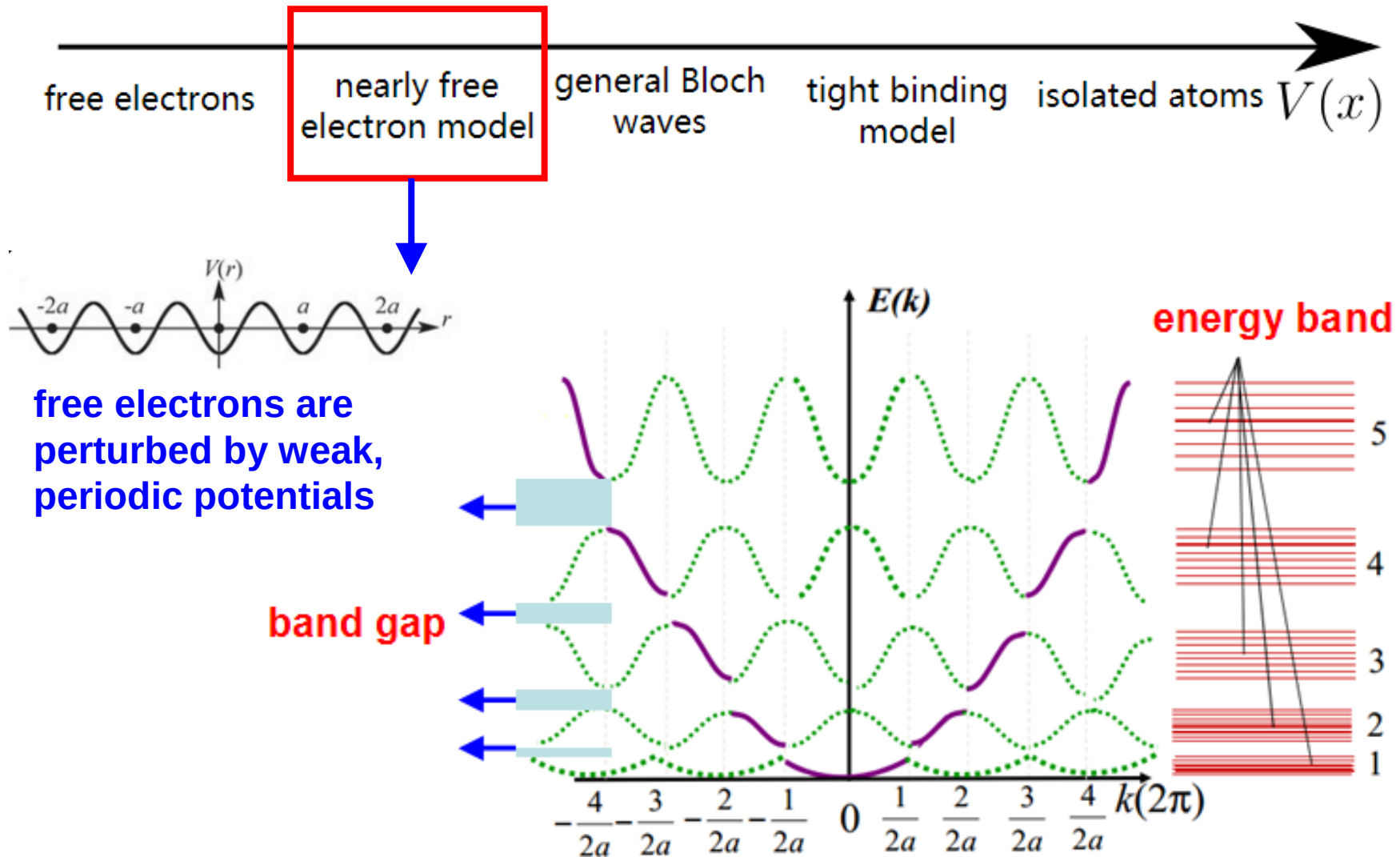


SiO₂

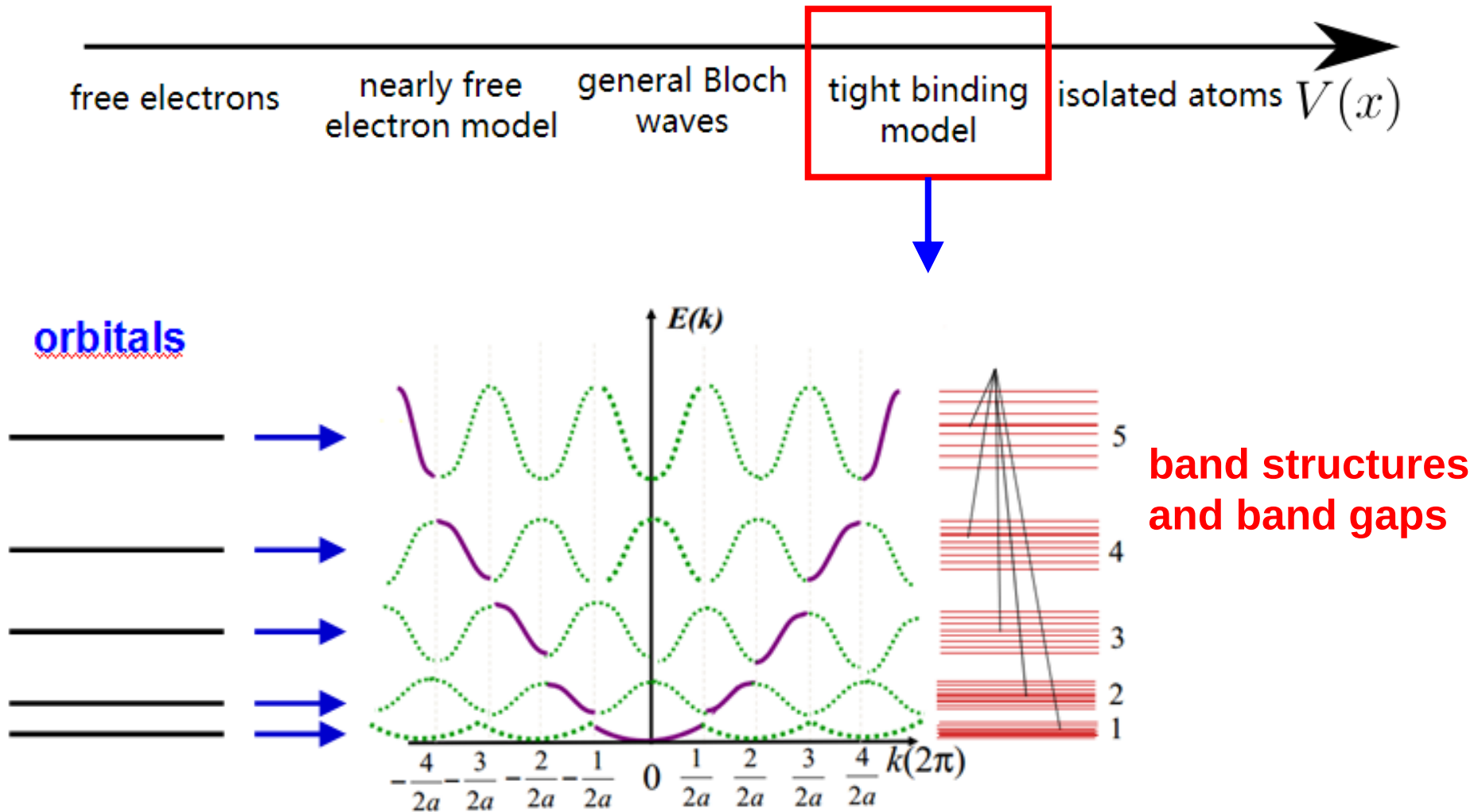


Silicon

Formation of Band Gaps

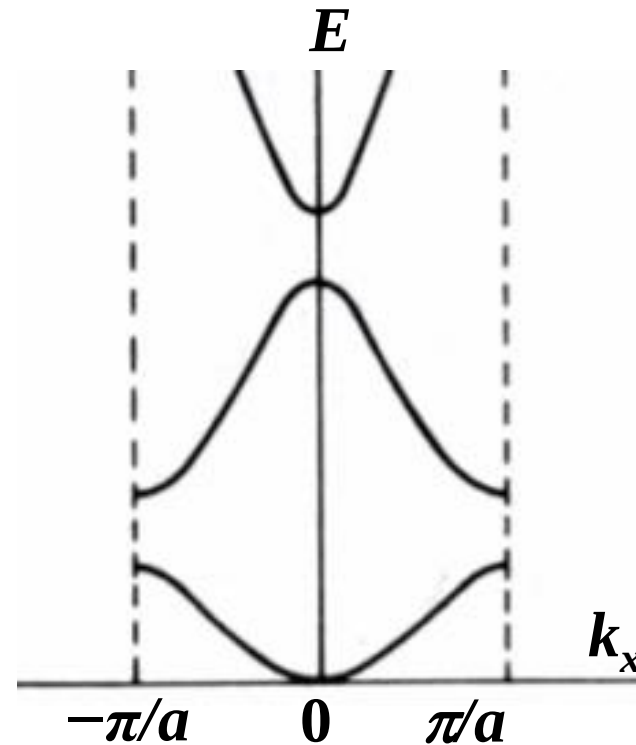
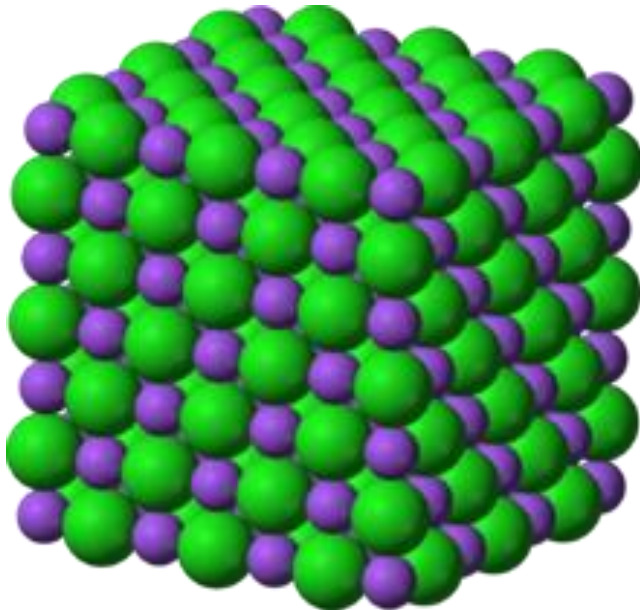


Formation of Band Gaps



discrete orbitals are combined to form quasi-continuous bands

Formation of Band Gaps



Fill electrons into these bands

Q: Is it a metal or insulator?

State vs. Electron

energy state / level / orbital
能态 / 能级 / 轨道



determined by space, lattice,
environments, ...

electron / phonon / ...
电子 / 声子 / ...



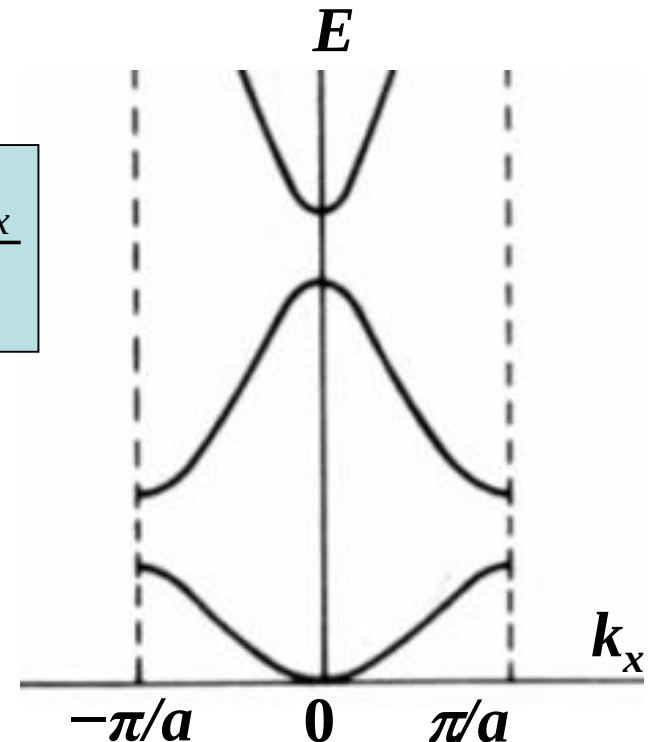
**There are infinite number
of states, but only finite
number of electrons.**

Energy States

How many energy states in each band?

$$\begin{aligned}
 &\text{number of states} \\
 &= 2 \cdot \frac{\text{size of FBZ}}{\text{density of } k \text{ points}} \\
 &= 2 \cdot \frac{2\pi / a}{2\pi / L_x} \\
 &= 2 \frac{L_x}{a} \\
 &= 2N
 \end{aligned}$$

$$k_x = \frac{2\pi n_x}{L_x}$$



FBZ - First Brillouin Zone

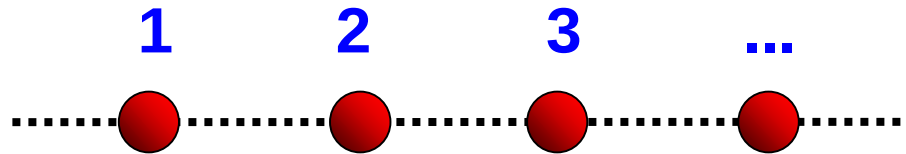
$N = L/a$ - number of primitive cells

2 - spin up and down

Q: How about 2D and 3D cases? 10

1D Chain of *Monovalent* Atoms

Each atom has *one* valence electron (Na, K, ...)

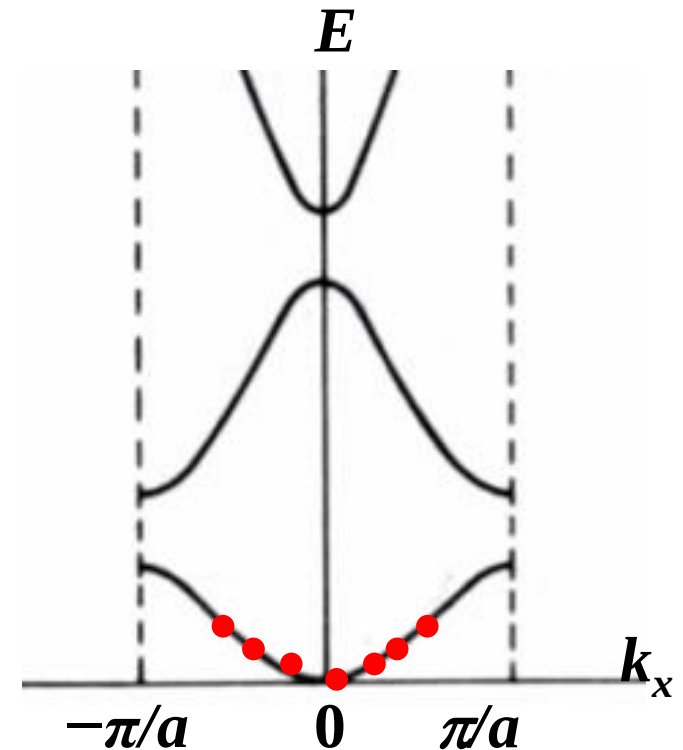


total number of states = $2N$

$$N = L/a$$

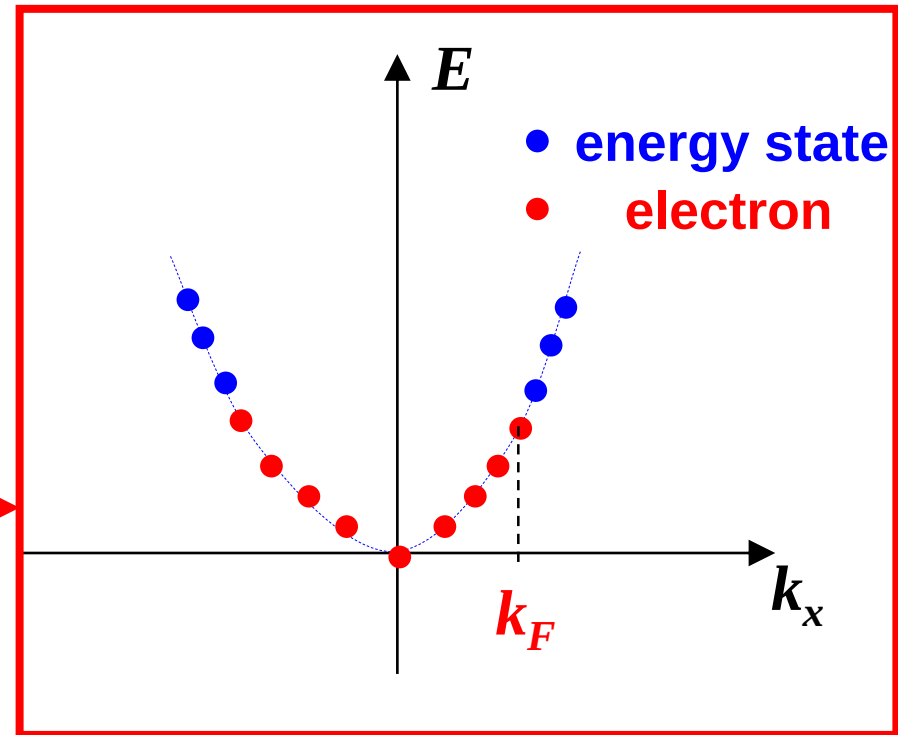
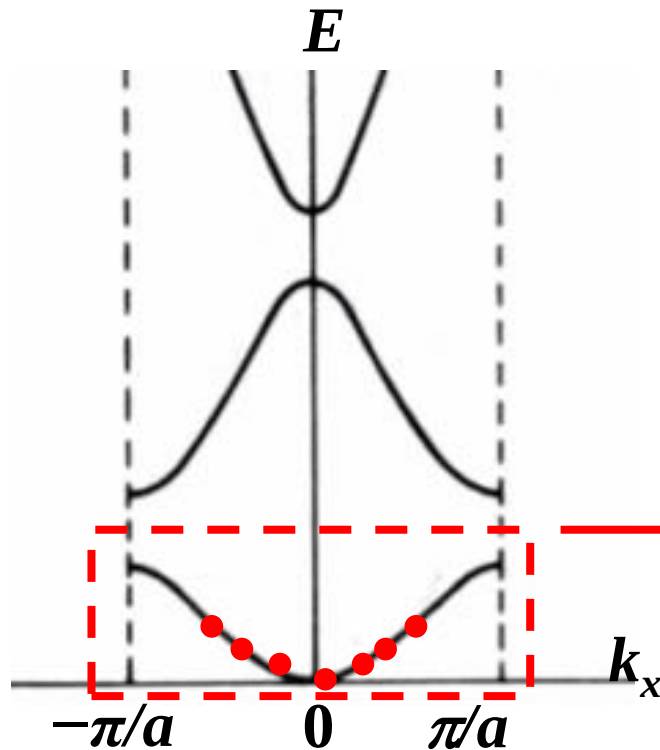
= number of primitive cells
= number of valence electrons

Electrons only fill half of the first band



• electron

1D Chain of *Monovalent* Atoms



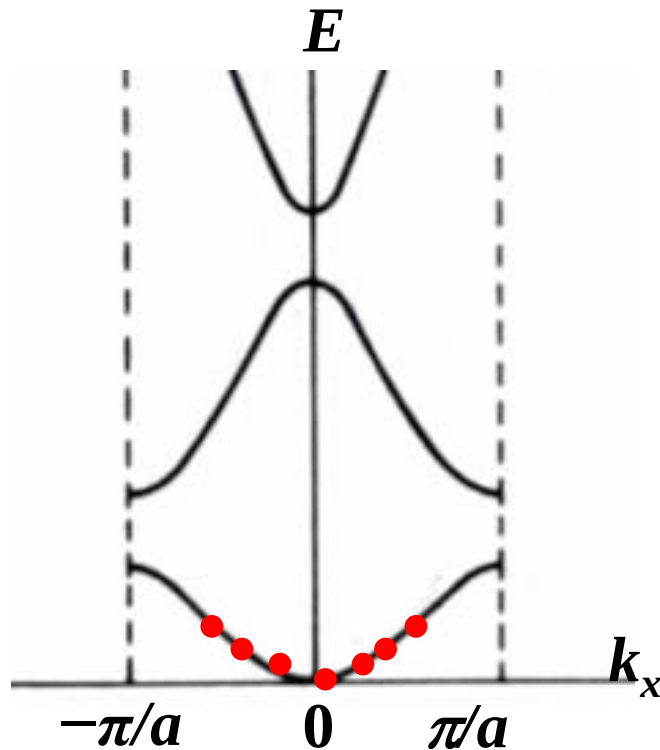
*Electrons only fill half
of the first band*

parabolic approximation

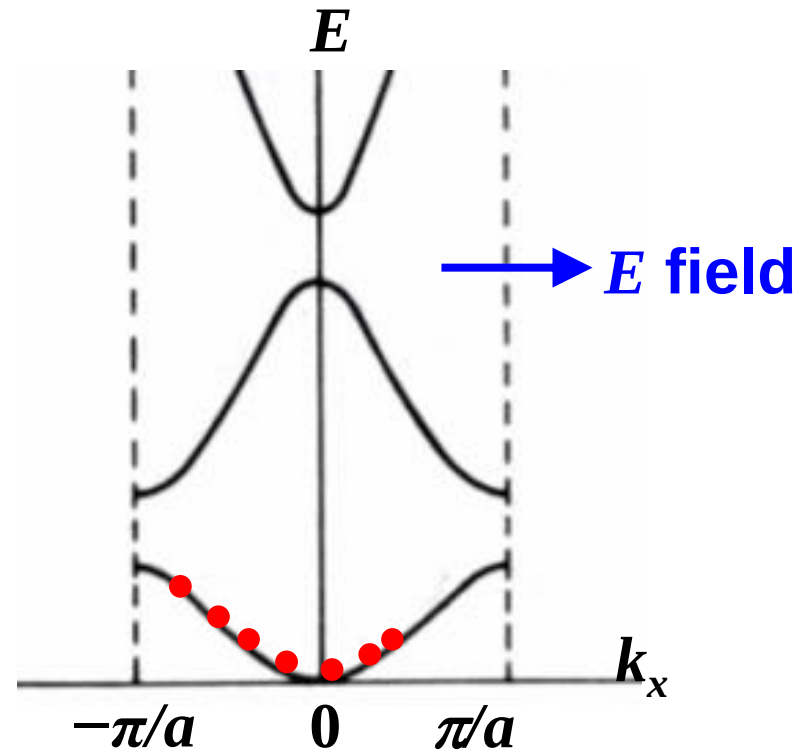
Free Electrons

Conductor

1D Chain of *Monovalent* Atoms



when $E = 0$, $v = 0$
no current



when $E \neq 0$, $v > 0$
electric current

$$\mathbf{j} = \sigma \mathbf{E}$$

1D Chain of *Monovalent* Atoms

Question:

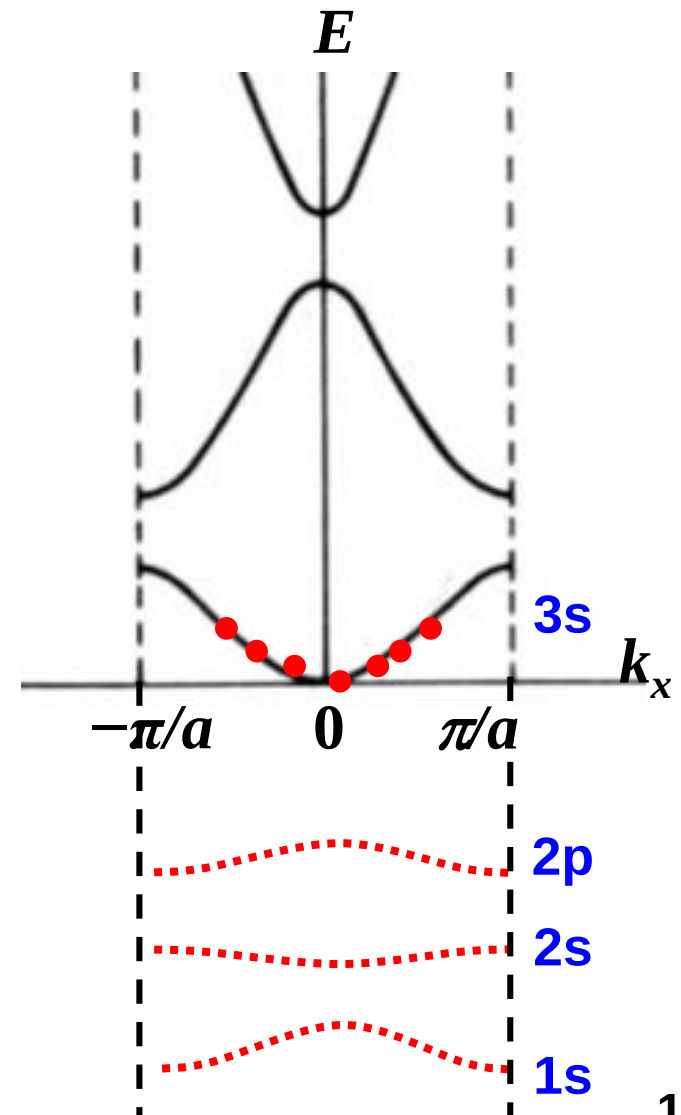
A Na atom has 11 electrons.

$[1s^2 2s^2 2p^6] 3s^1$

Where are the rest 10 electrons?

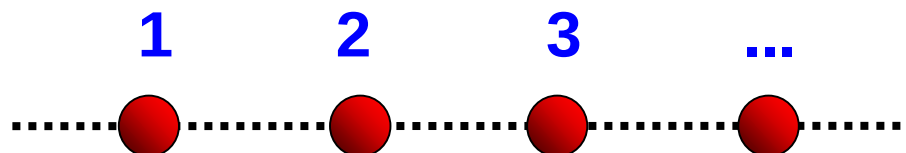
Answer:

Localized electrons have lower energies. All the lower bands are completely full. So they have no contribution to conductivity.



1D Chain of *Divalent* Atoms

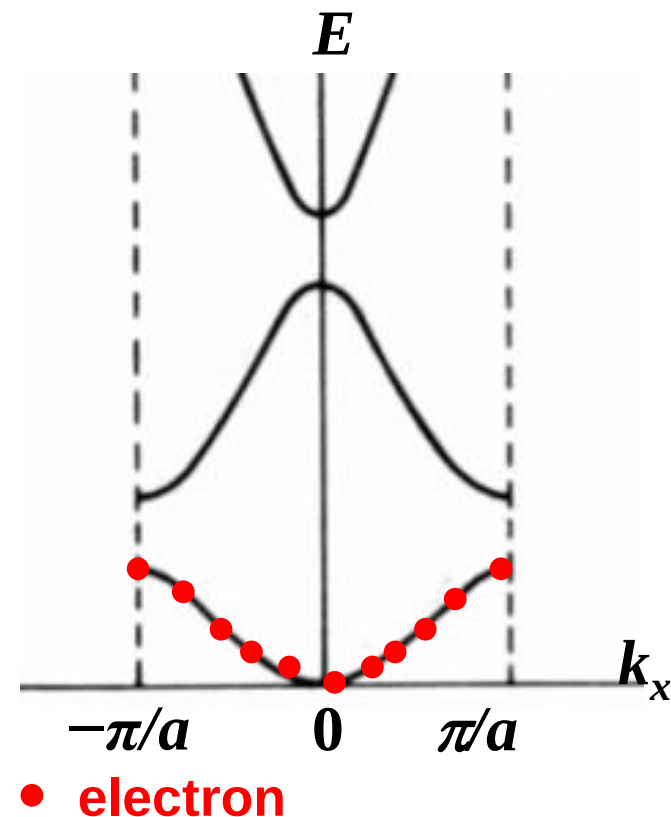
Each atom has *two* valence electrons (Mg, Ca, ...)



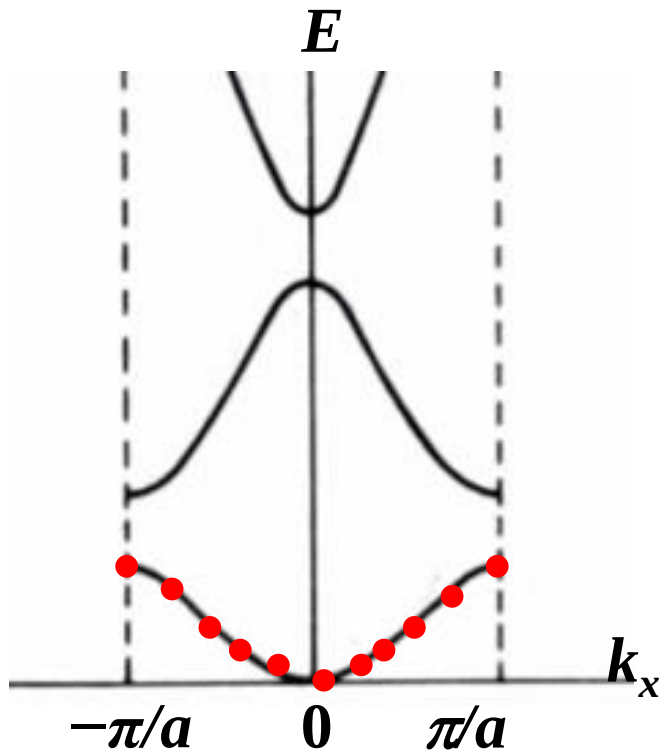
total number of states = $2N$

number of valence electrons
= $2L/a = 2N$

Electrons fill the entire band

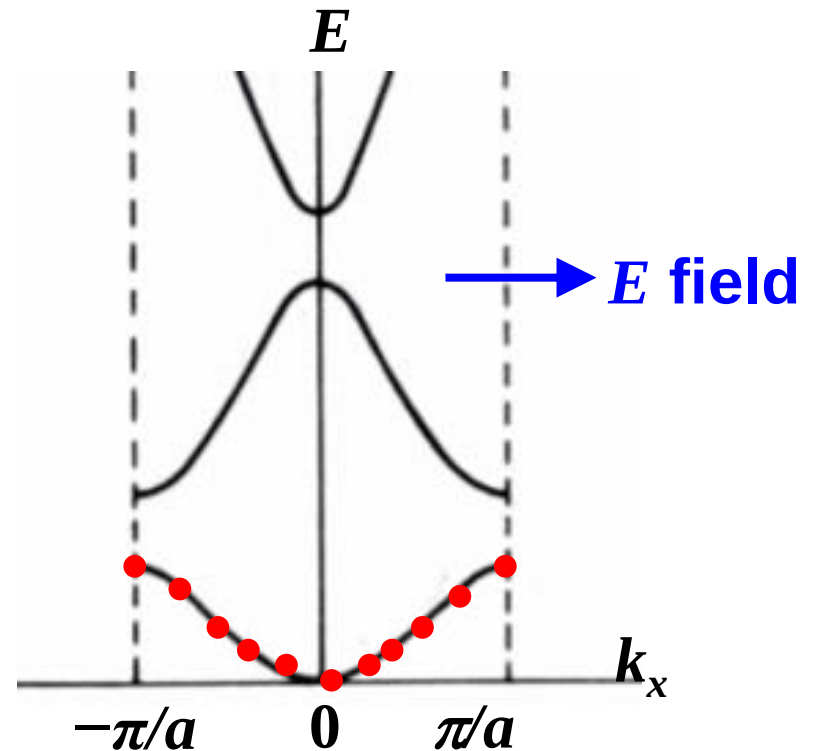


1D Chain of *Divalent* Atoms



when $E = 0$, $v = 0$
no current

Electrons fill the entire band →

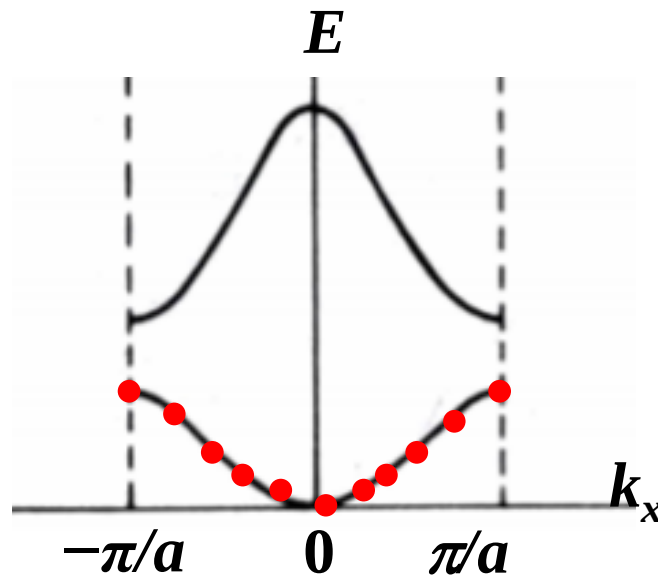


when $E \neq 0$, $v = 0$
no current

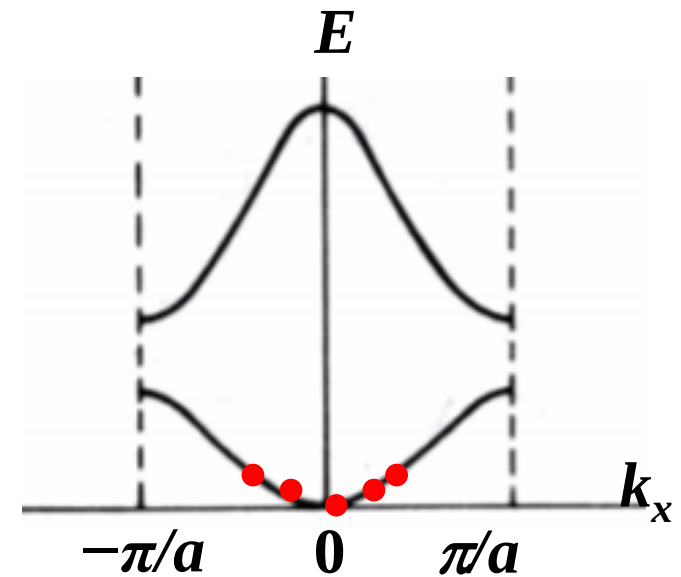
Insulator
(at $T = 0$ K)

Metal vs. Insulator

- Metals and Insulators are fundamentally different
 - Insulators are not just “bad” conductors
 - An insulator has all bands completely filled or empty
 - A metal has partially filled bands



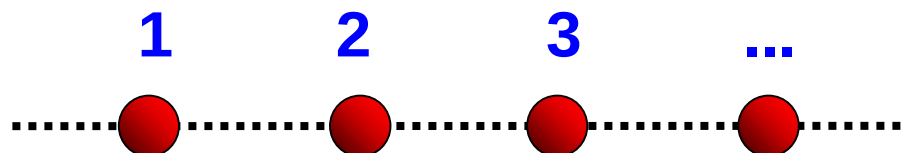
Insulator



Metal

1D Chain of *Trivalent* Atoms

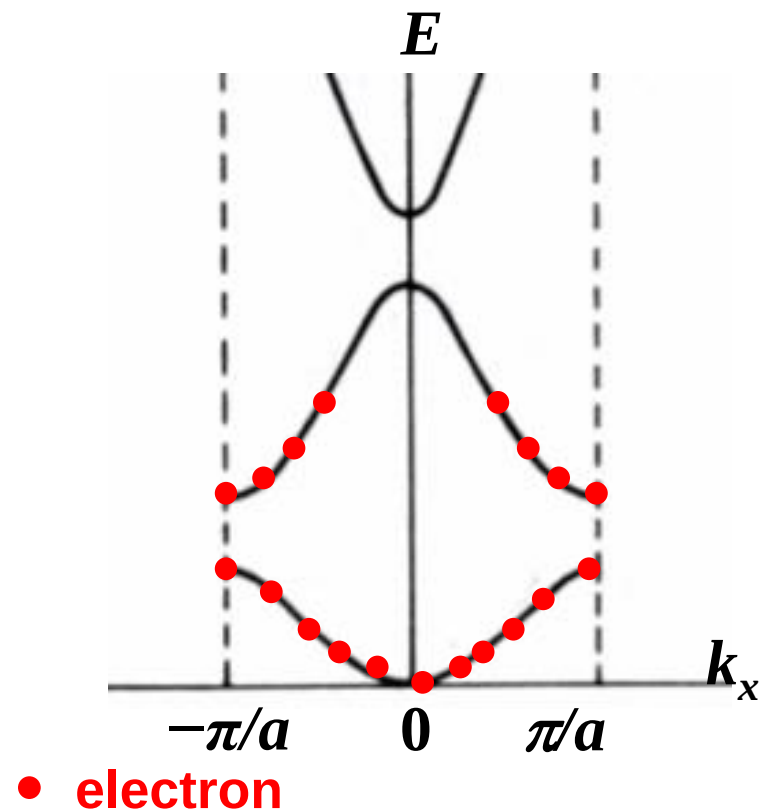
Each atom has *three* valence electrons (Al, Ga, ...)



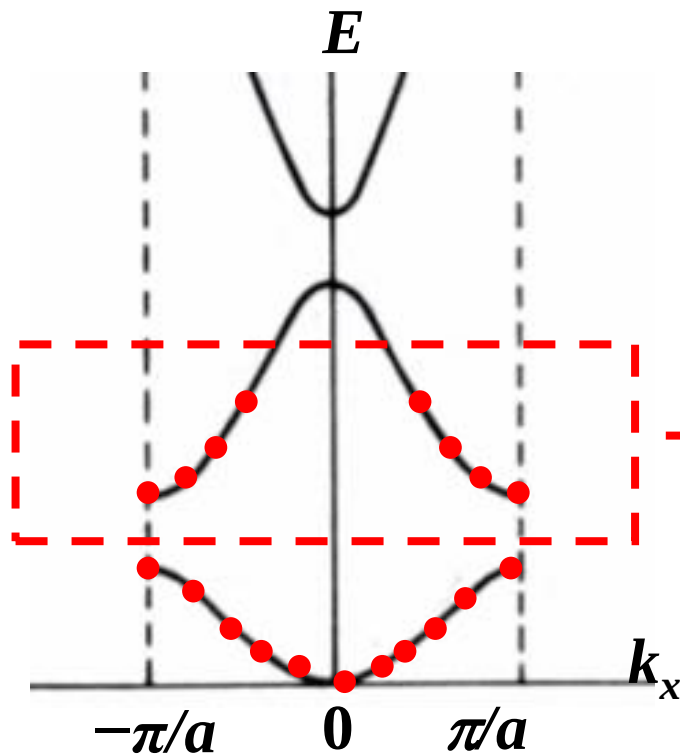
total number of states = $2N$

number of valence electrons
= $3L/a = 3N$

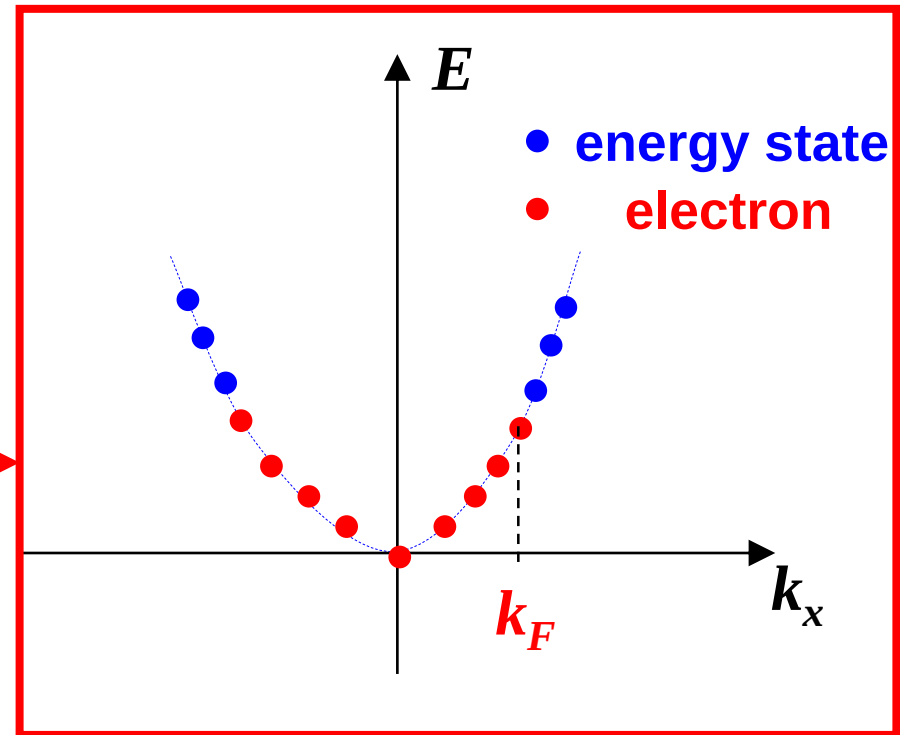
*Electrons start to fill the
second band*



1D Chain of *Trivalent* Atoms



• electron

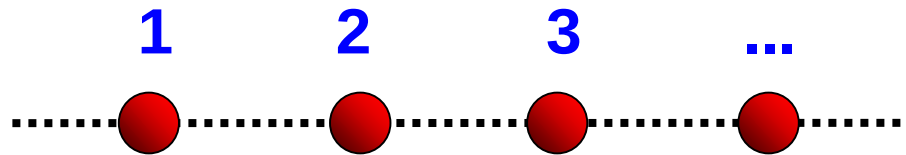


Free Electrons

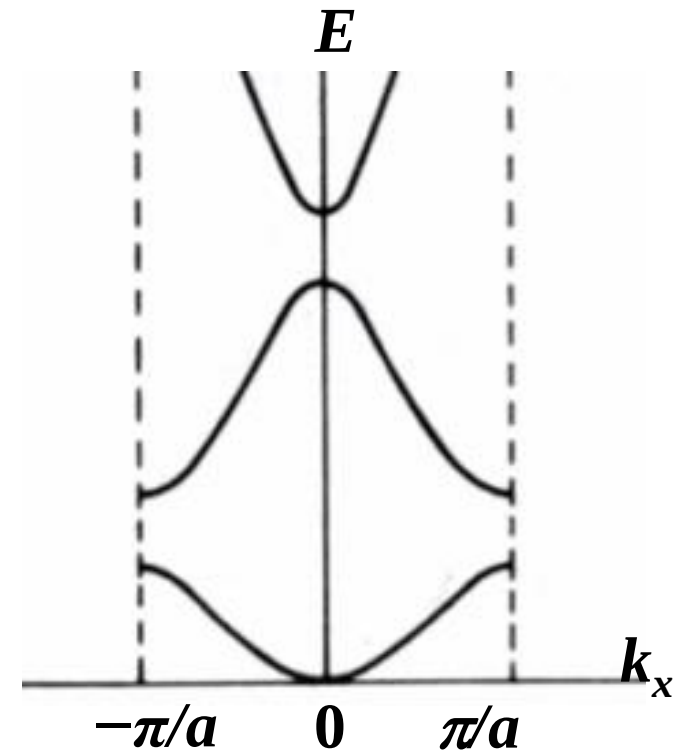
Conductor

1D Chain of *Quadrivalent* Atoms

Each atom has *four* valence electrons (C, Si, ...)



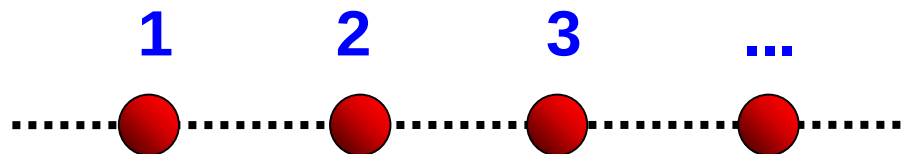
*Q: How do electrons fill the bands?
Is it a conductor, or an insulator?*



• electron

1D Chain of *Divalent* Atoms

Each atom has *two* valence electrons (Mg, Ca, ...)

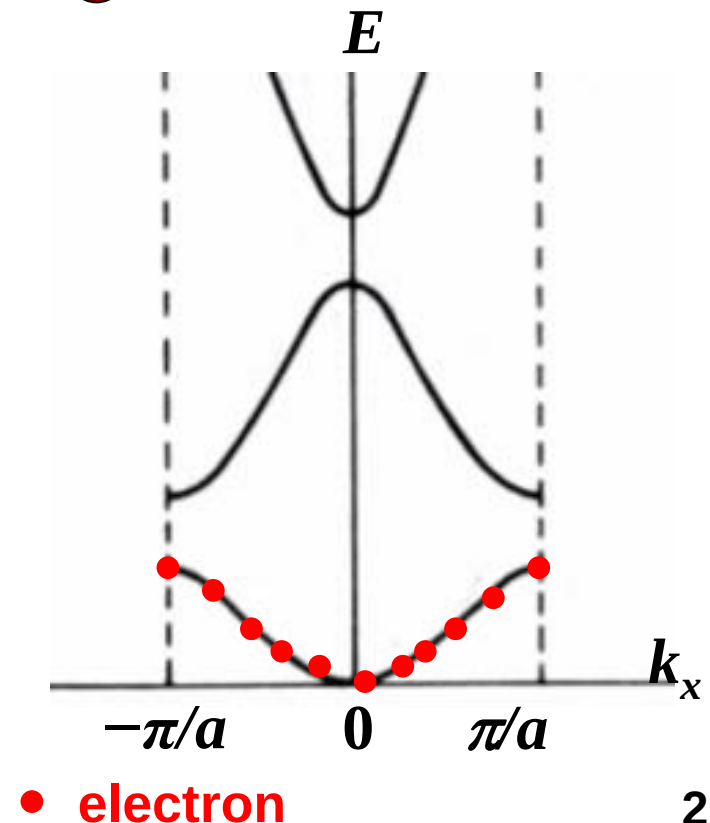


Electrons fill the entire band

--->

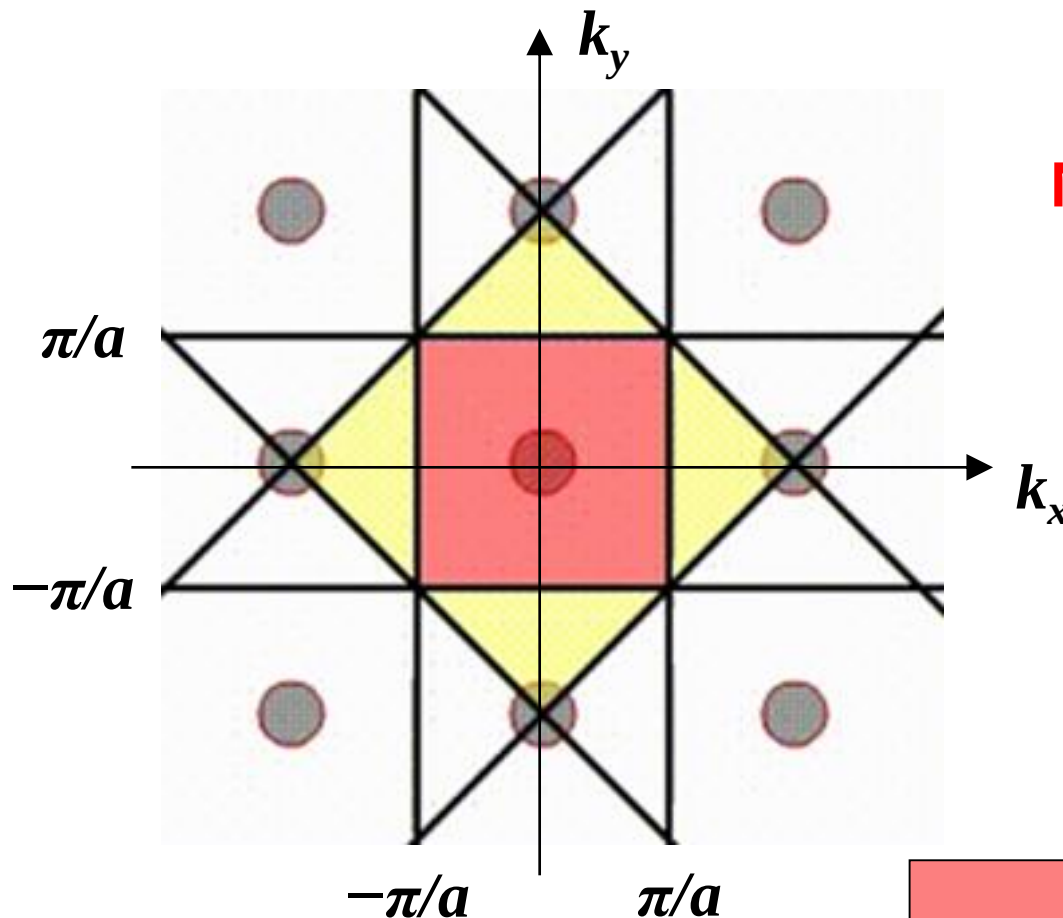
insulator

Q: But we know divalent metals like Mg and Ca are conductors, why?



2D case

Assume a square lattice in the reciprocal space



Number of states in each
Brillouin zone

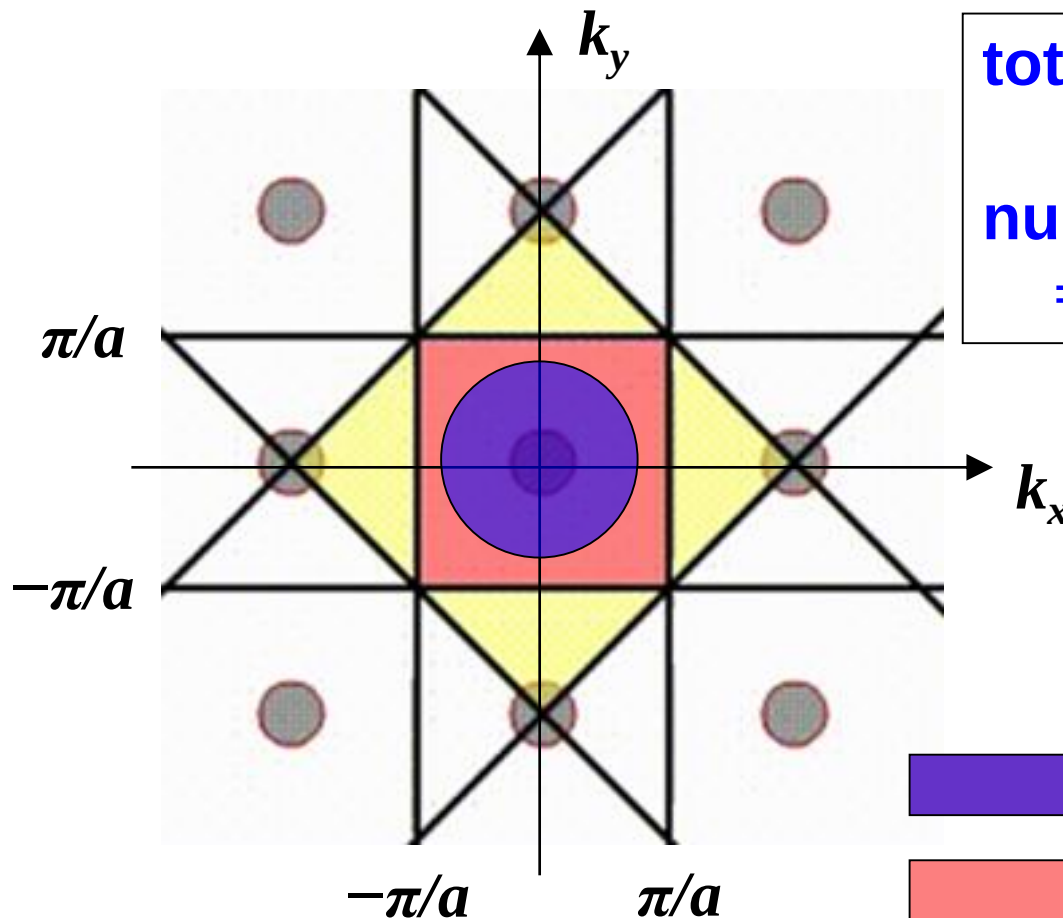
total number of states
 $= 2N$

First BZ

Second BZ

2D case of *Monovalent* Atoms

Each atom has *one* valence electrons (Na, K, ...)



total number of states = $2N$

number of valence electrons
= $A/a/a = N$

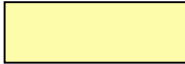
FBZ is not full



Conductor

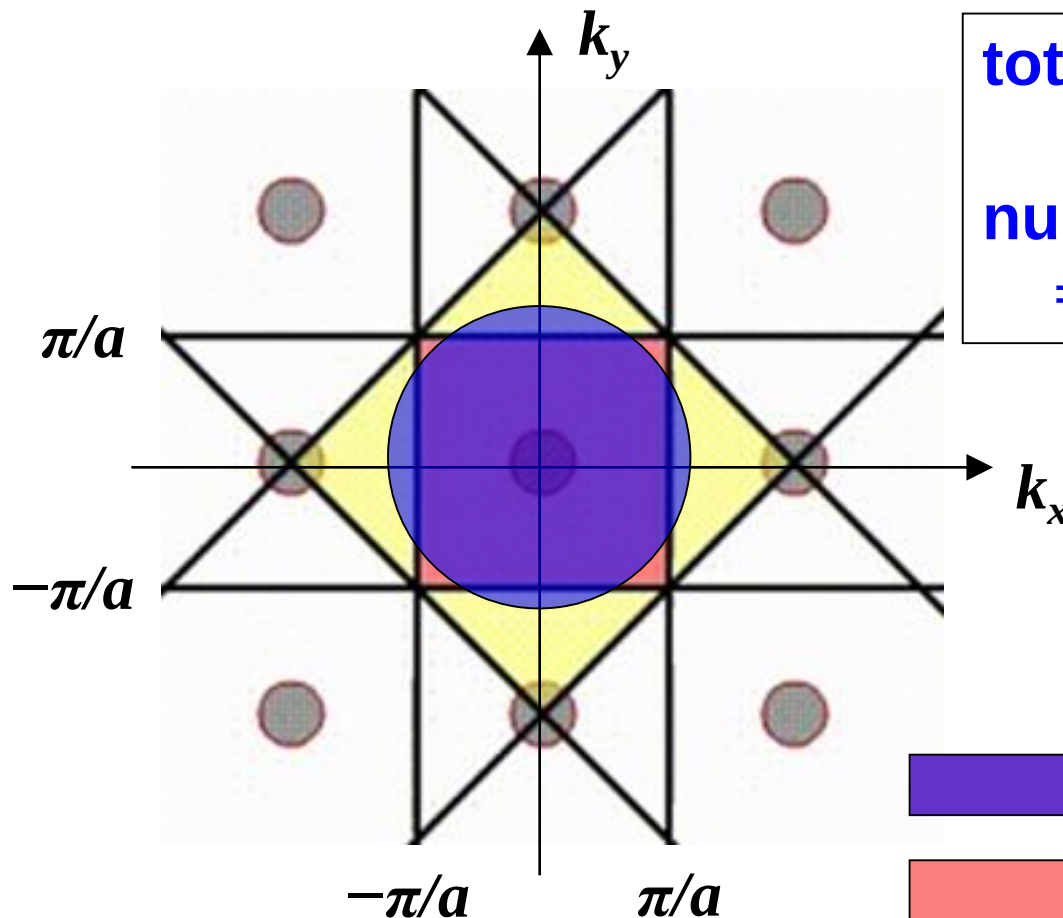
 **electrons**

 **First BZ**

 **Second BZ**

2D case of *Divalent* Atoms

Each atom has *two* valence electrons (Mg, Ca, ...)



total number of states = $2N$

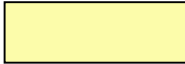
number of valence electrons
= $2A/a/a = 2N$

*start to fill SBZ, but
FBZ is still not full*

Conductor (semimetal)

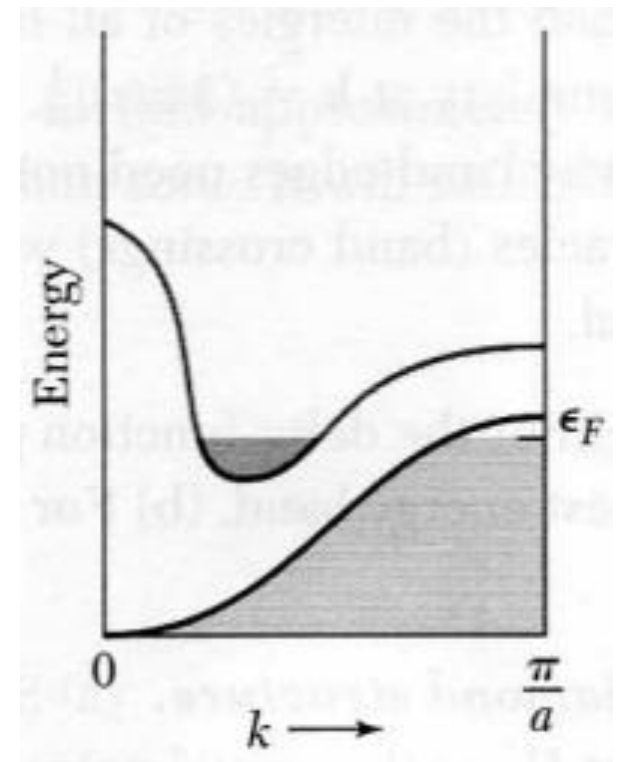
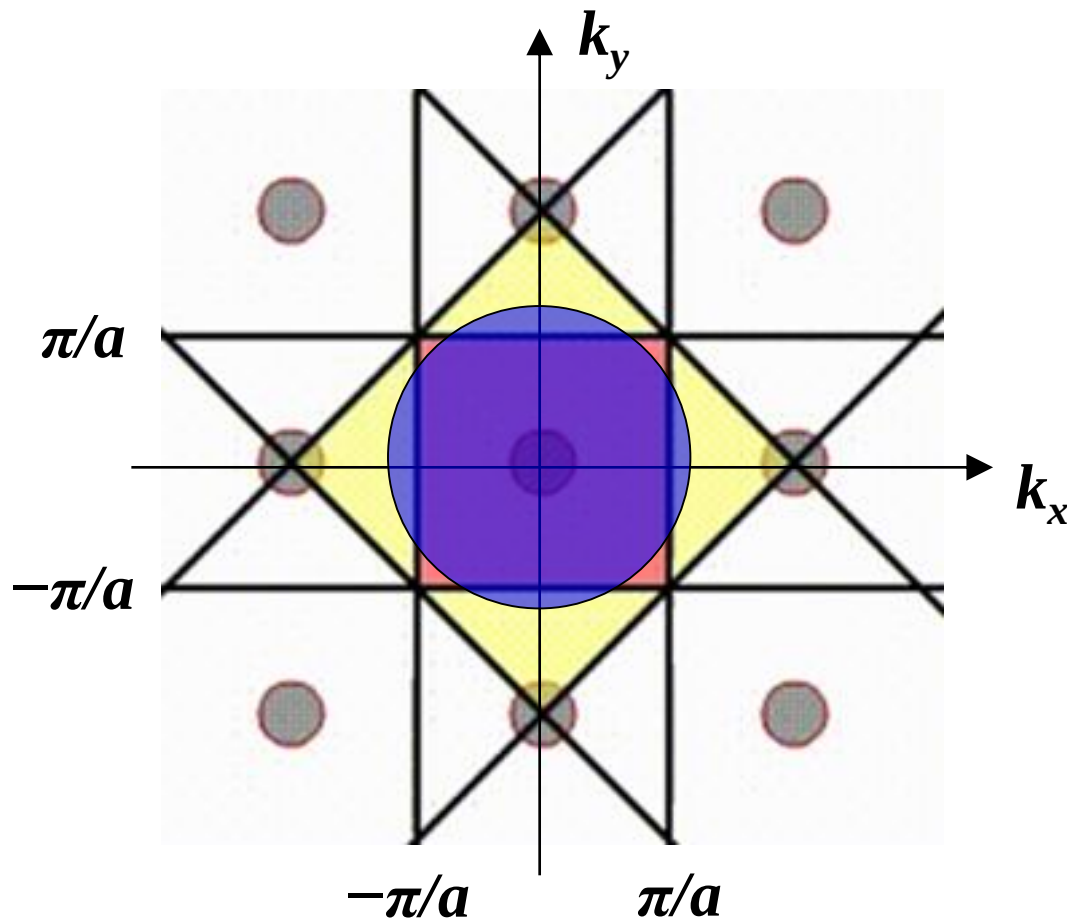
 **electrons**

 **First BZ**

 **Second BZ**

2D case of *Divalent* Atoms

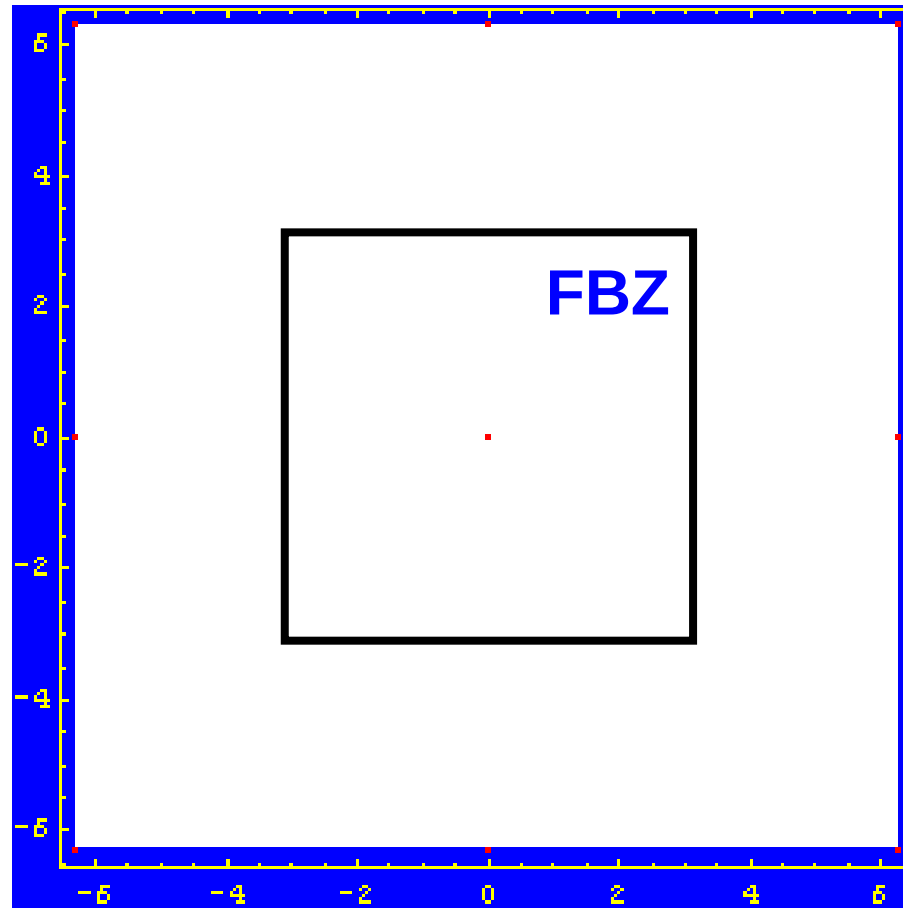
Each atom has *two* valence electrons (Mg, Ca, ...)



semimetal 半金属

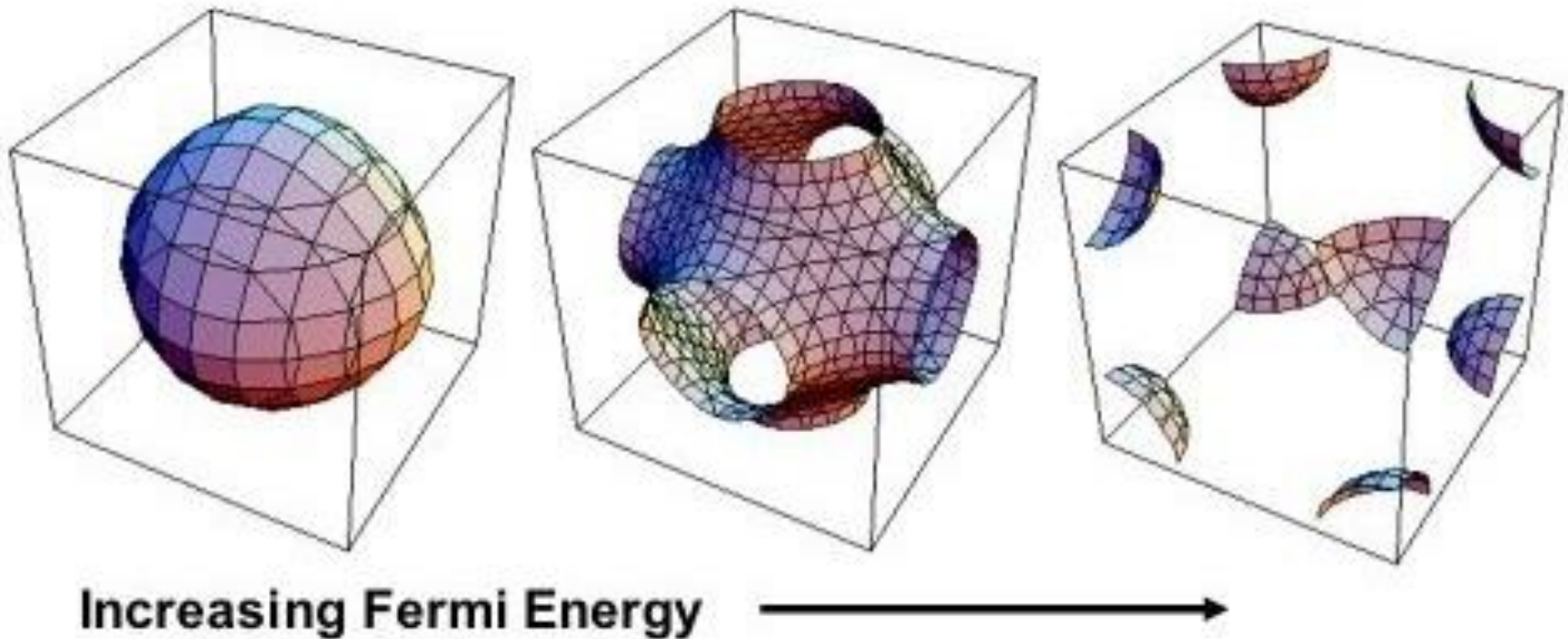
2D Fermi Surface

simple square



3D Fermi Surface

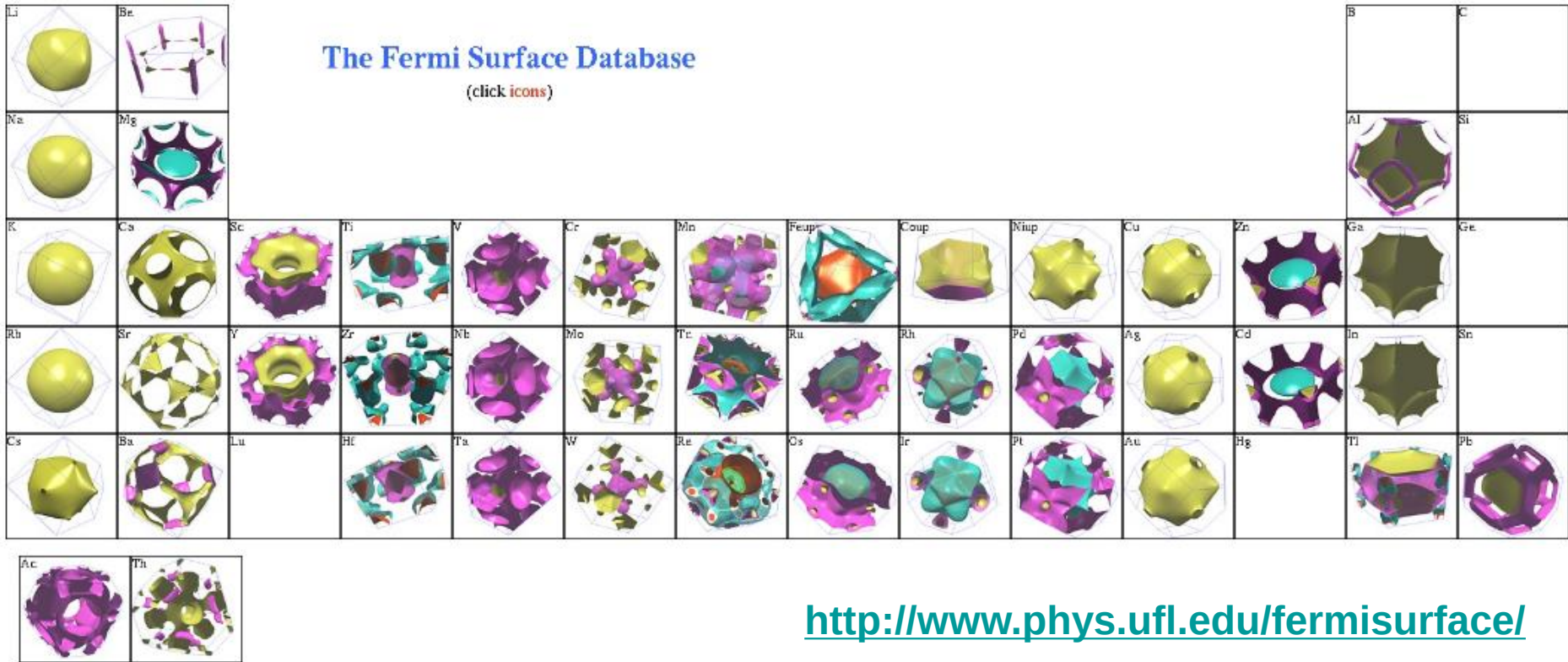
simple cubic



generated with a tight binding model:

<http://home.cc.umanitoba.ca/~loly/fermisurf2.html>

3D Fermi Surface – More Examples

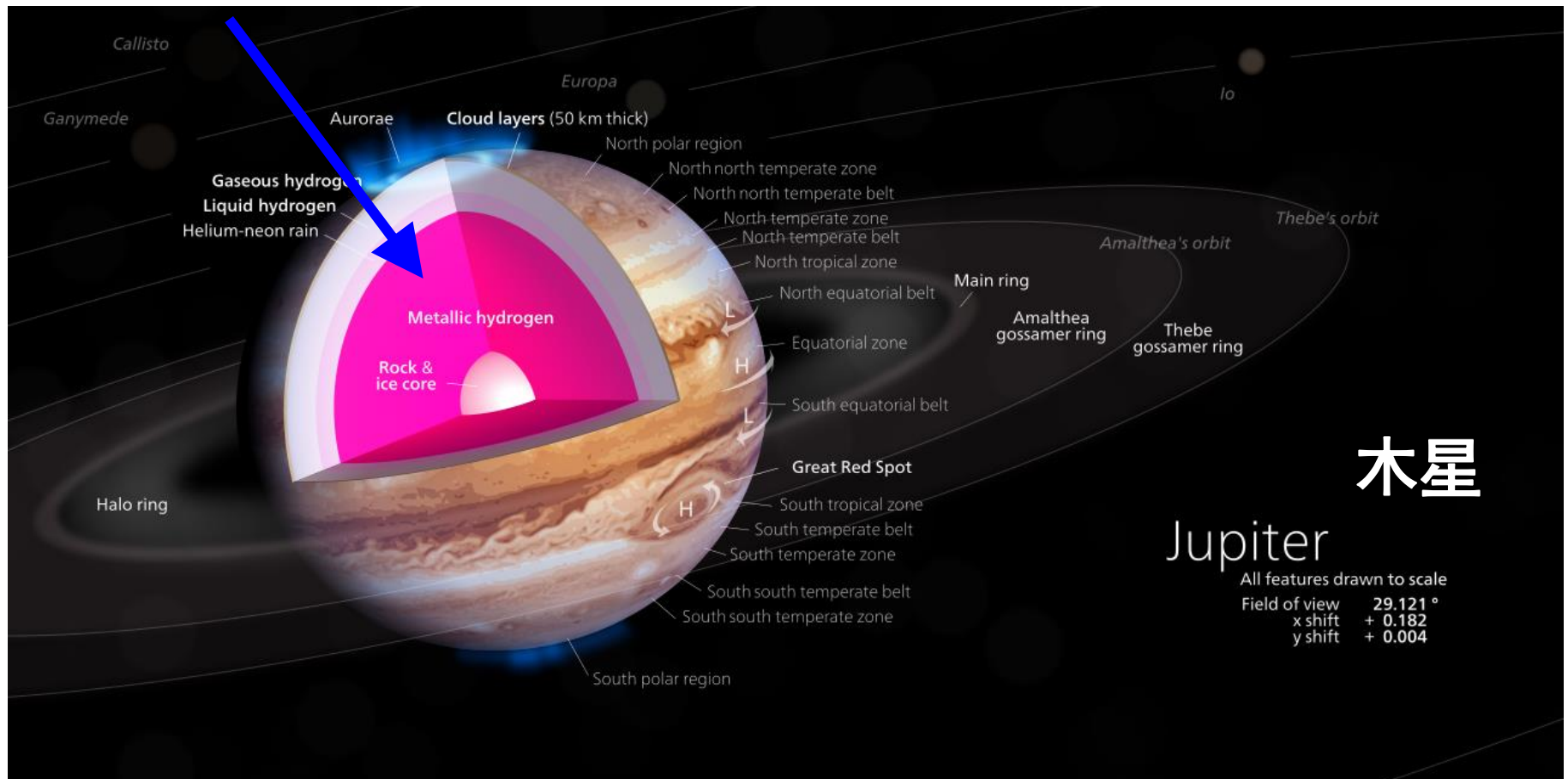


Fermi surface is the surface in reciprocal space which separates occupied from unoccupied electron states at $T = 0$ K.

https://www.angelo.edu/faculty/kboudrea/periodic/physical_metals.htm

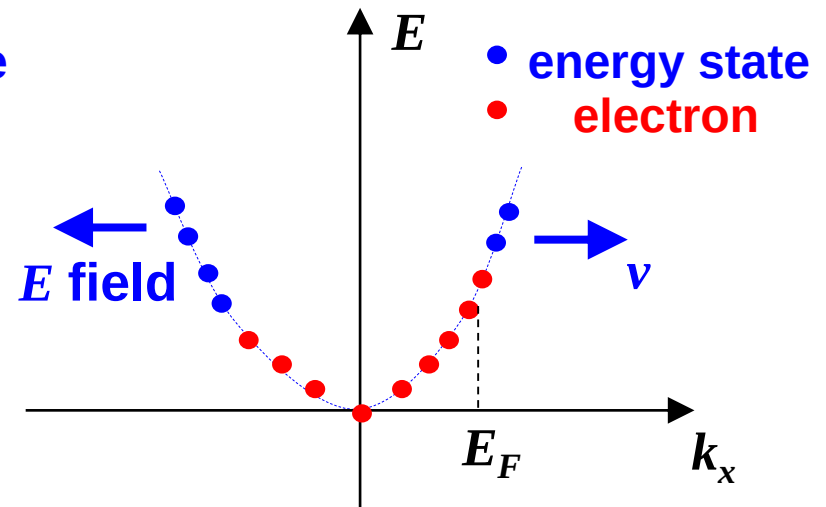
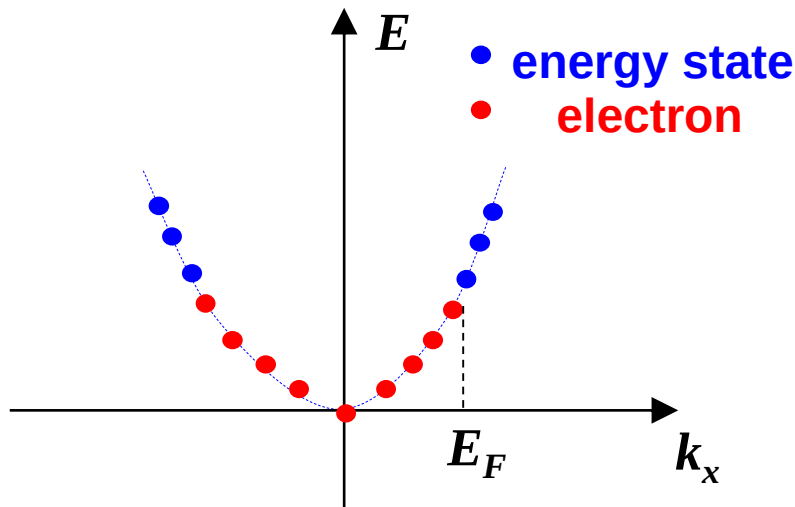
Solid Hydrogen is also Metallic

Metallic hydrogen only exists under very high pressure



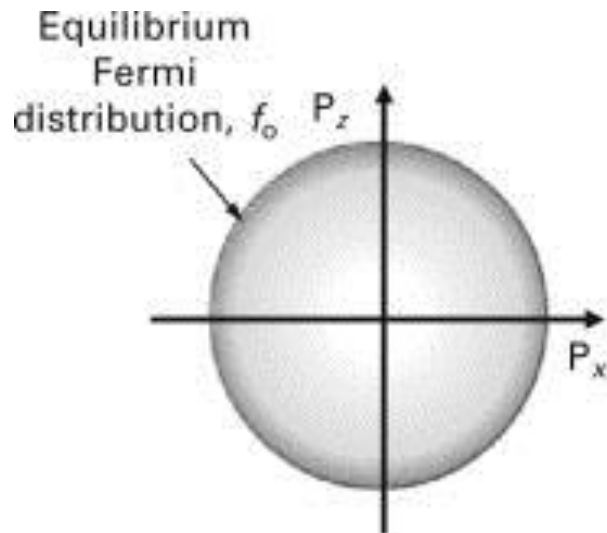
Electrical Conductivity - Revisit

1D

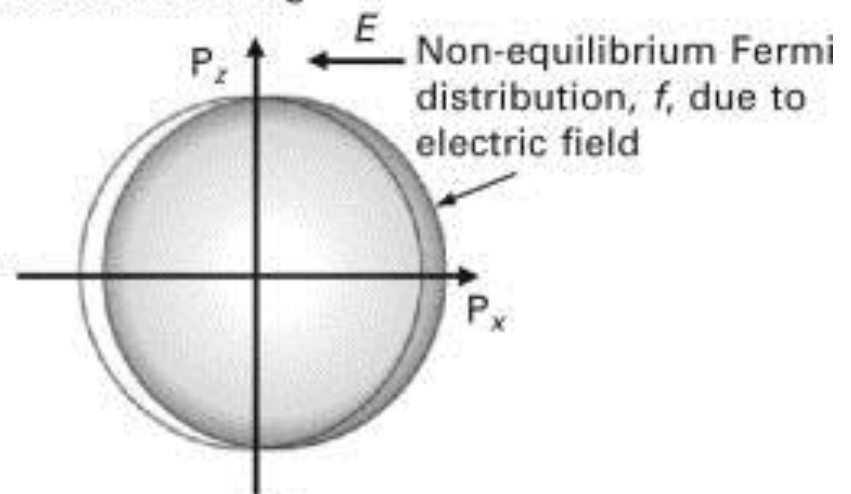


2D

3D



Diffuse scattering

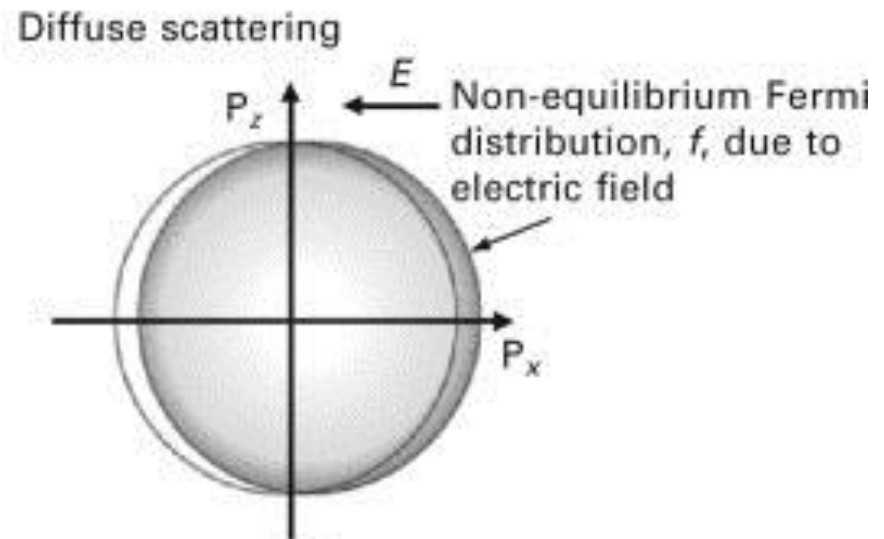
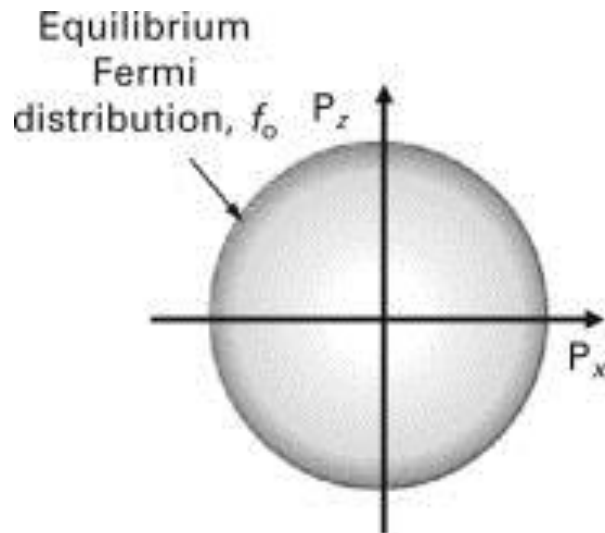


Electrical Conductivity - Revisit

- Only electrons **near the Fermi surface** contribute to electrical (and thermal) conductivity in metals

$$\sigma = e^2 \cdot \tau(E_F) \cdot \int_{\text{occupied levels}} \frac{2d\mathbf{k}}{(2\pi)^3} \cdot \frac{1}{M^*(\mathbf{k})}$$

Ashcroft & Mermin, Chap.13

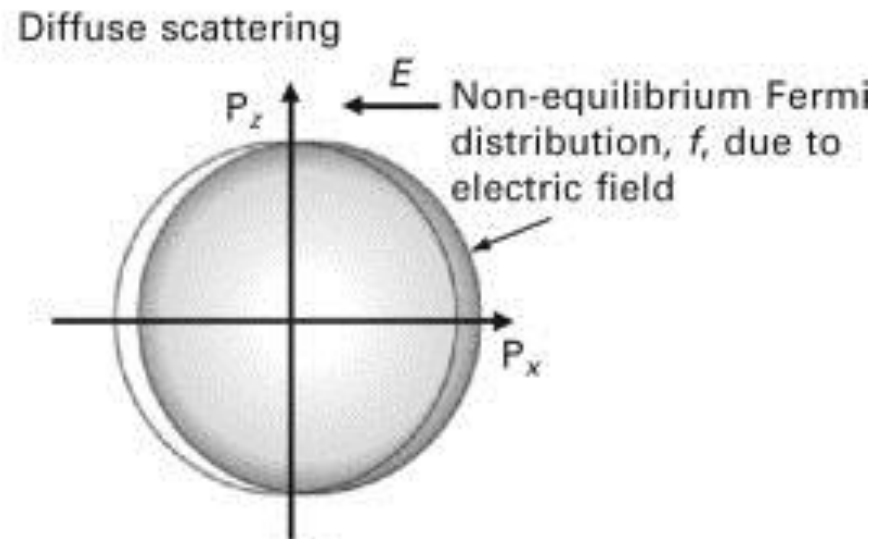
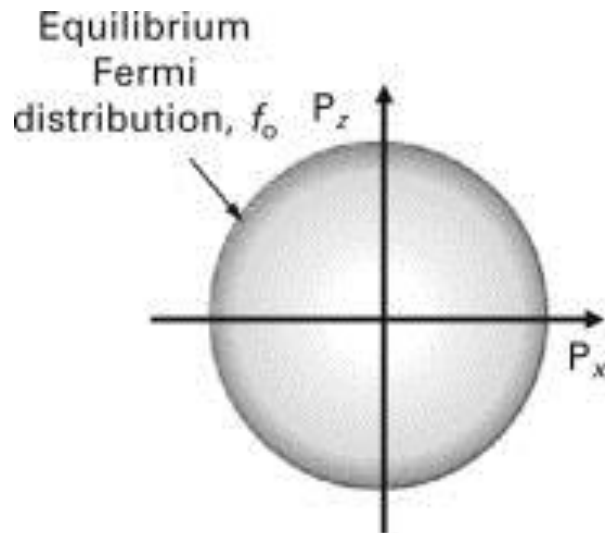


Electrical Conductivity - Revisit

- Only electrons **near the Fermi surface** contribute to electrical (and thermal) conductivity in metals

compare

$$\sigma = ne \frac{v}{E} = ne\mu = \frac{ne^2\tau}{m^*}$$



Electrical Conductivity - Revisit

- Only electrons **near the Fermi surface** contribute to electrical (and thermal) conductivity in metals



Electrical Conductivity - Revisit

$$\sigma = ne \frac{v}{E} = ne\mu = \frac{ne^2\tau}{m^*}$$

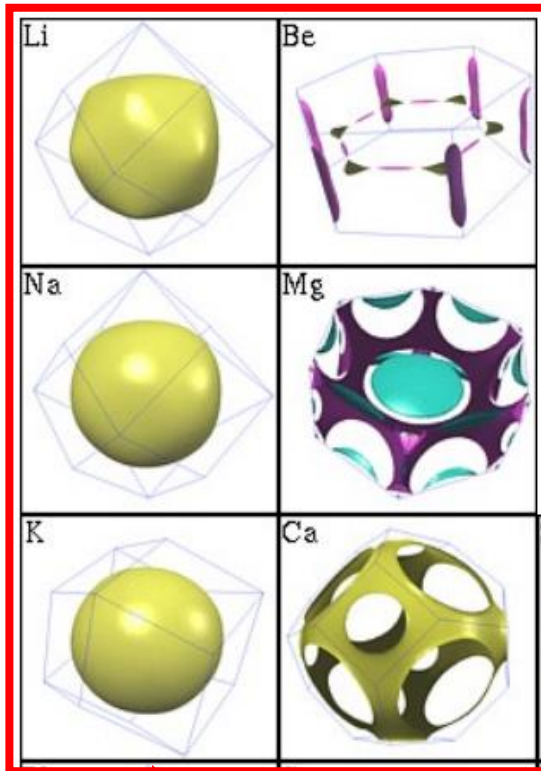
		Classical	Quantum
electron density	n	all valence electrons	electrons near Fermi surface (depend on E field)
μ	τ		
velocity	v	average (depend on temperature)	Fermi velocity v_F
mass	m	free electron m_0	effective mass m^*

3D Fermi Surface

IA

IIA

Metals in Group IA and IIA have very different properties



Periodic Table of the Elements

1 IA	2 IIA											13 IIIA	14 IVA	15 VA	16 VIA	17 VIIA	18 VIIIA
1 H Hydrogen 1.008																	2 He Helium 4.003
3 Li Lithium 6.941	4 Be Beryllium 9.012											5 B Boron 10.811	6 C Carbon 12.011	7 N Nitrogen 14.007	8 O Oxygen 15.999	9 F Fluorine 18.998	10 Ne Neon 20.180
11 Na Sodium 22.990	12 Mg Magnesium 24.305											13 Al Aluminum 26.982	14 Si Silicon 28.086	15 P Phosphorus 30.974	16 S Sulfur 32.06	17 Cl Chlorine 35.453	18 Ar Argon 39.948
19 K Potassium 39.098	20 Ca Calcium 40.078	21 Sc Scandium 44.956	22 Ti Titanium 47.867	23 V Vanadium 50.942	24 Cr Chromium 51.996	25 Mn Manganese 54.938	26 Fe Iron 55.845	27 Co Cobalt 58.933	28 Ni Nickel 58.693	29 Cu Copper 63.546	30 Zn Zinc 65.38	31 Ga Gallium 69.723	32 Ge Germanium 72.63	33 As Arsenic 74.922	34 Se Selenium 78.971	35 Br Bromine 79.904	36 Kr Krypton 84.798
37 Rb Rubidium 84.468	38 Sr Strontium 87.62	39 Y Yttrium 88.906	40 Zr Zirconium 91.224	41 Nb Niobium 92.906	42 Mo Molybdenum 95.95	43 Tc Technetium 98.907	44 Ru Ruthenium 101.07	45 Rh Rhodium 102.906	46 Pd Palladium 106.42	47 Ag Silver 107.868	48 Cd Cadmium 112.411	49 In Indium 114.818	50 Sn Tin 118.710	51 Sb Antimony 121.760	52 Te Tellurium 127.6	53 I Iodine 126.904	54 Xe Xenon 131.29
55 Cs Cesium 132.905	56 Ba Barium 137.328	57-71 Lanthanide Series	72 Hf Hafnium 178.49	73 Ta Tantalum 180.945	74 W Tungsten 183.84	75 Re Rhenium 186.207	76 Os Osmium 190.23	77 Ir Iridium 192.217	78 Pt Platinum 195.085	79 Au Gold 196.967	80 Hg Mercury 200.592	81 Tl Thallium 204.383	82 Pb Lead 207.2	83 Bi Bismuth 208.980	84 Po Polonium [209]	85 At Astatine [210]	86 Rn Radon 222.018
87 Fr Francium 223.020	88 Ra Radium 226.025	89-103 Actinide Series	104 Rf Rutherfordium [261]	105 Db Dubnium [262]	106 Sg Seaborgium [266]	107 Bh Bohrium [264]	108 Hs Hassium [269]	109 Mt Meitnerium [268]	110 Ds Darmstadtium [289]	111 Rg Roentgenium [272]	112 Cn Copernicium [277]	113 Nh Nihonium [284]	114 Fl Flerovium [289]	115 Mc Moscovium [288]	116 Lv Livermorium [293]	117 Ts Tennessine [294]	118 Uuo Unbinilium [294]
57 La Lanthanum 138.905	58 Ce Cerium 140.116	59 Pr Praseodymium 140.908	60 Nd Neodymium 144.243	61 Pm Promethium 144.913	62 Sm Samarium 150.36	63 Eu Europium 151.964	64 Gd Gadolinium 157.25	65 Tb Terbium 158.925	66 Dy Dysprosium 162.500	67 Ho Holmium 164.930	68 Er Erbium 167.259	69 Tm Thulium 168.934	70 Yb Ytterbium 173.055	71 Lu Lutetium 174.967			
89 Ac Actinium 227.028	90 Th Thorium 232.038	91 Pa Protactinium 231.036	92 U Uranium 238.029	93 Np Neptunium 237.048	94 Pu Plutonium 244.064	95 Am Americium 243.061	96 Cm Curium 247.070	97 Bk Berkelium 247.070	98 Cf Californium 251.080	99 Es Einsteinium [254]	100 Fm Fermium [257]	101 Md Mendelevium [258]	102 No Nobelium [259]	103 Lr Lawrencium [262]			

Legend:

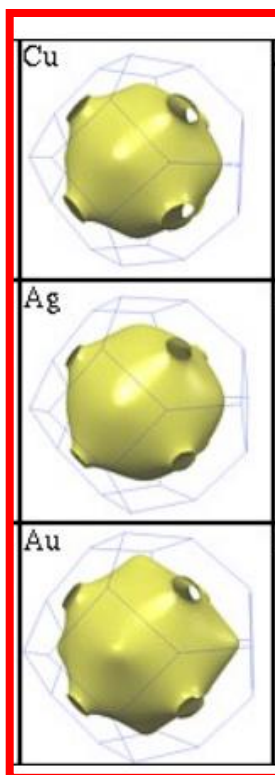
- Alkali Metal
- Alkaline Earth
- Transition Metal
- Basic Metal
- Semimetal
- Nonmetal
- Halogen
- Noble Gas
- Lanthanide
- Actinide

spherical shape
close to free electrons

3D Fermi Surface

IB

Metals in Group IB are very good conductors



Periodic Table of the Elements

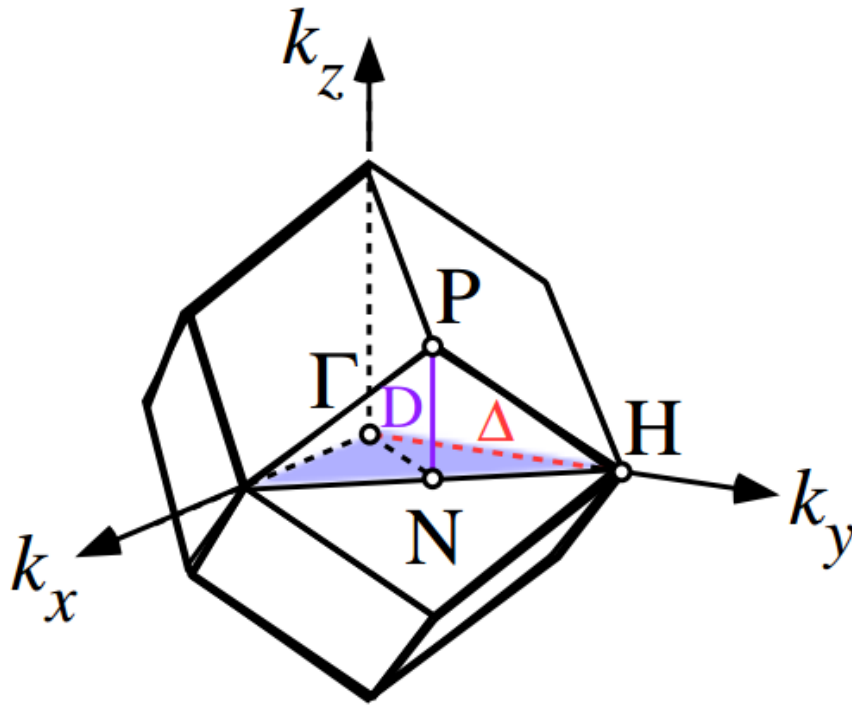
1 IA	2 IIA											13 IIIA	14 IVA	15 VA	16 VIA	17 VIIA	18 VIIIA
1 H 1.008												5 B 10.811	6 C 12.011	7 N 14.007	8 O 15.999	9 F 18.998	10 Ne 20.180
3 Li 6.941	4 Be 9.012											13 Al 26.982	14 Si 28.086	15 P 30.974	16 S 32.06	17 Cl 35.453	18 Ar 39.948
11 Na 22.99	12 Mg 24.305	21 Sc 44.955	22 Ti 47.867	23 V 50.942	24 Cr 51.996	25 Mn 54.938	26 Fe 55.845	27 Co 58.933	28 Ni 58.693	29 Cu 63.546	30 Zn 65.38	31 Ga 69.723	32 Ge 72.63	33 As 74.922	34 Se 78.971	35 Br 79.904	36 Kr 83.798
19 K 39.098	20 Ca 40.078	39 Y 88.906	40 Zr 91.224	41 Nb 92.905	42 Mo 95.94	43 Tc 98.907	44 Ru 101.07	45 Rh 102.905	46 Pd 106.42	47 Ag 107.868	48 Cd 112.411	49 In 114.818	50 Sn 118.710	51 Sb 121.760	52 Te 127.6	53 I 126.904	54 Xe 131.29
37 Rb 84.468	38 Sr 87.62	57-71 Lanthanide Series	72 Hf 178.49	73 Ta 180.948	74 W 183.84	75 Re 186.207	76 Os 190.23	77 Ir 192.227	78 Pt 195.084	79 Au 196.967	80 Hg 200.592	81 Tl 204.383	82 Pb 207.2	83 Bi 208.980	84 Po [209]	85 At [210]	86 Rn [222]
55 Cs 132.905	56 Ba 137.327	89-103 Actinide Series	104 Rf [261]	105 Db [262]	106 Sg [266]	107 Bh [264]	108 Hs [269]	109 Mt [268]	110 Ds [271]	111 Rg [272]	112 Cn [277]	113 Nh [284]	114 Fl [289]	115 Mc [288]	116 Lv [293]	117 Ts [294]	118 Og [294]
87 Fr 223.020	88 Ra 226.025																

Legend: Alkali Metal, Alkaline Earth, Transition Metal, Basic Metal, Semimetal, Nonmetal, Halogen, Noble Gas, Lanthanide, Actinide

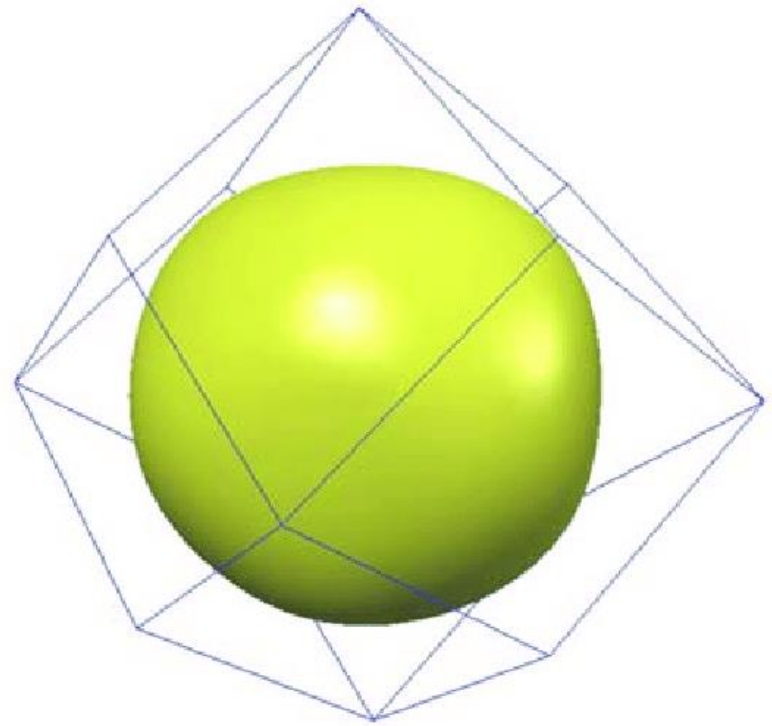
**spherical shape
close to free electrons**

3D case

sodium (BCC)



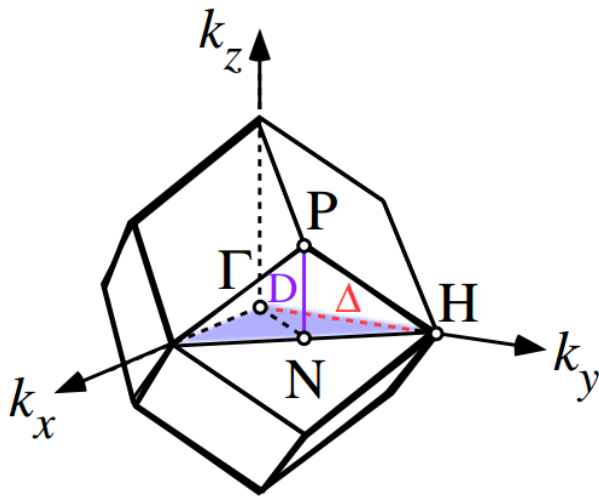
First BZ



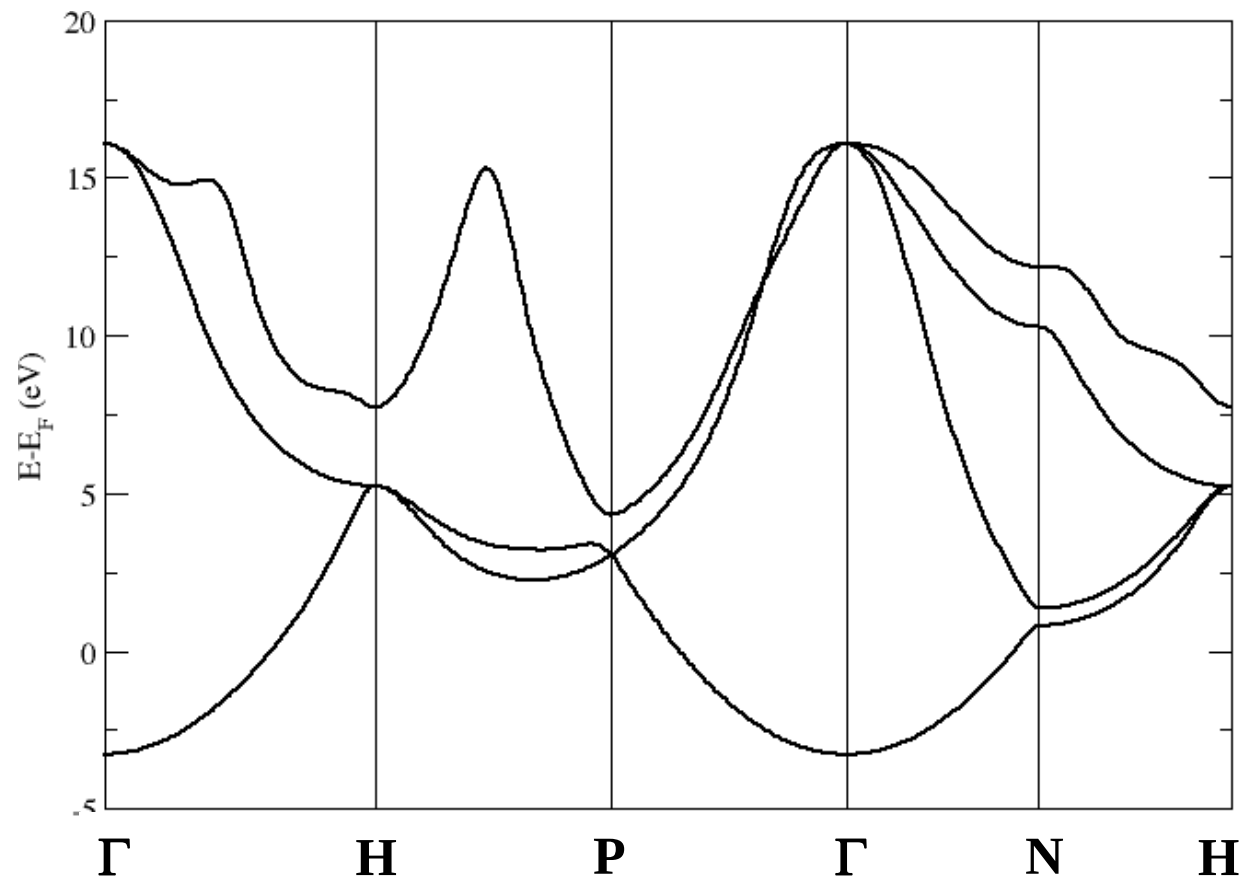
Fermi Surface

3D case

sodium (BCC): band structure

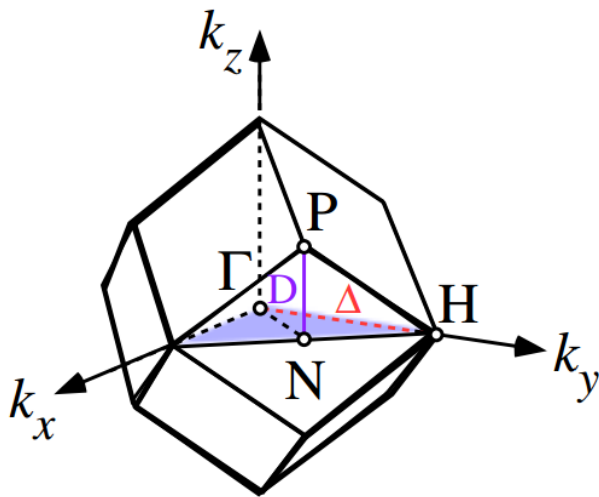


First BZ

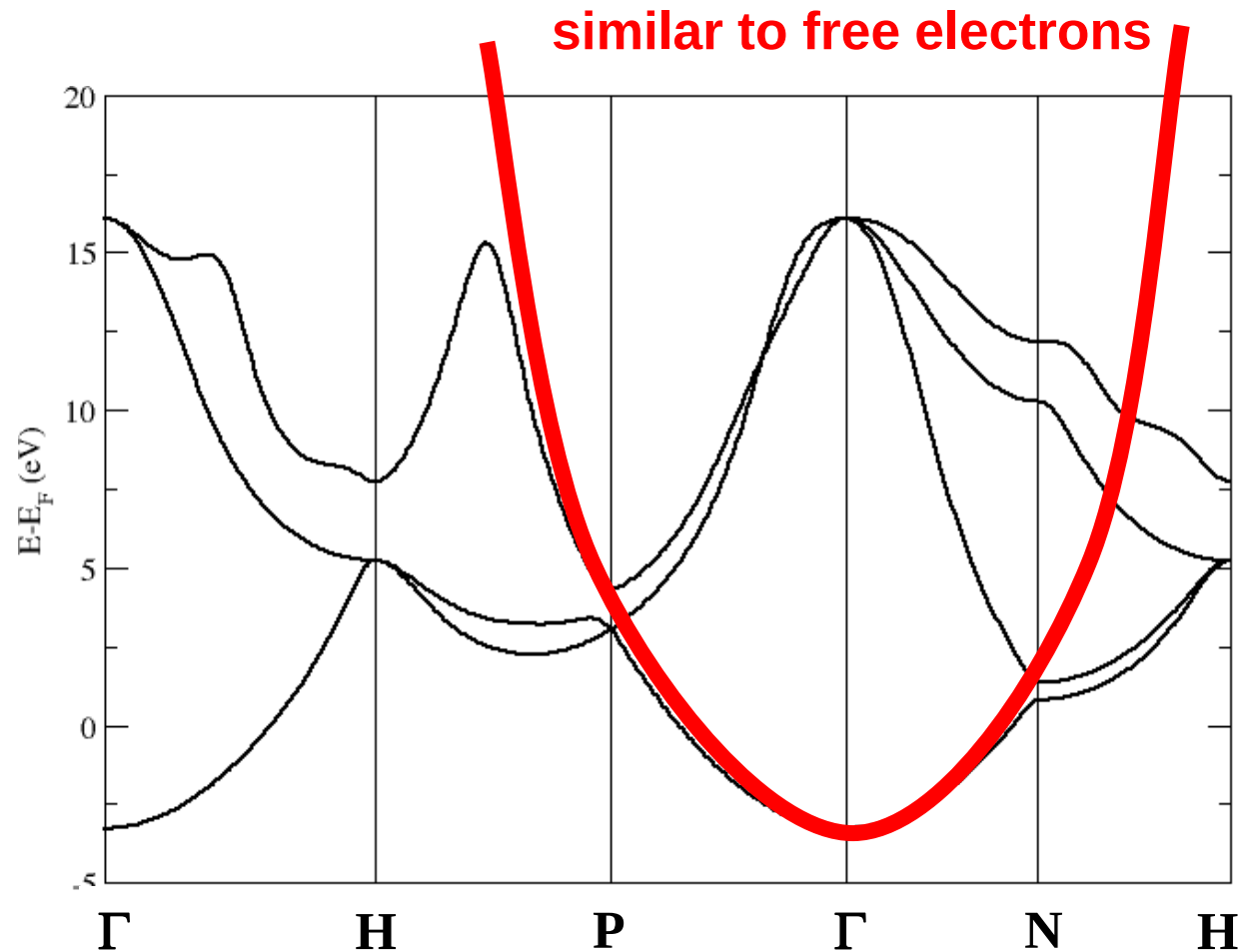


3D case

sodium (BCC): band structure

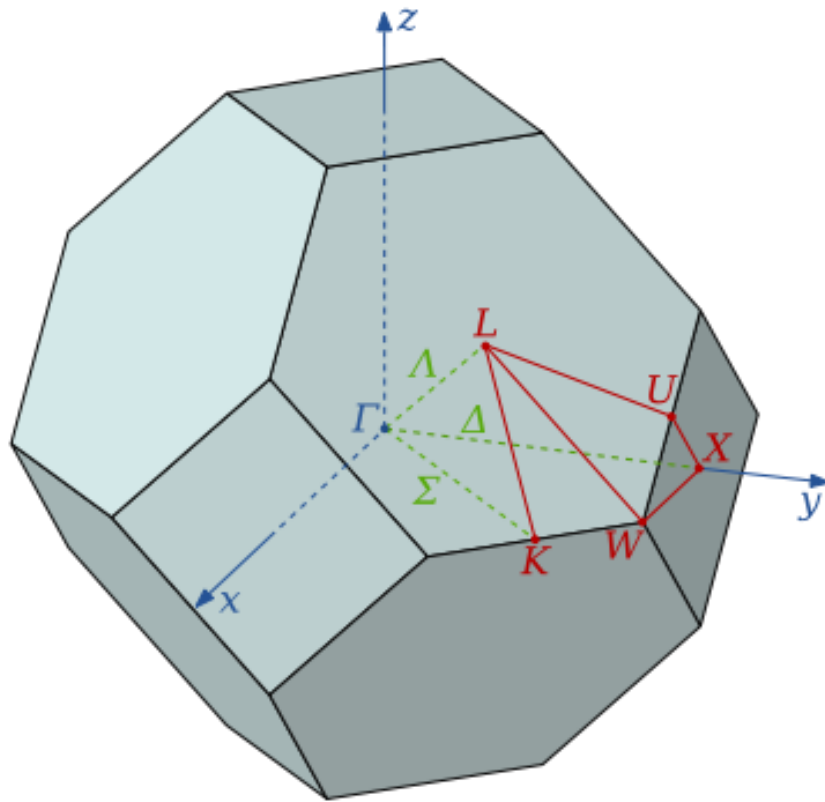


First BZ

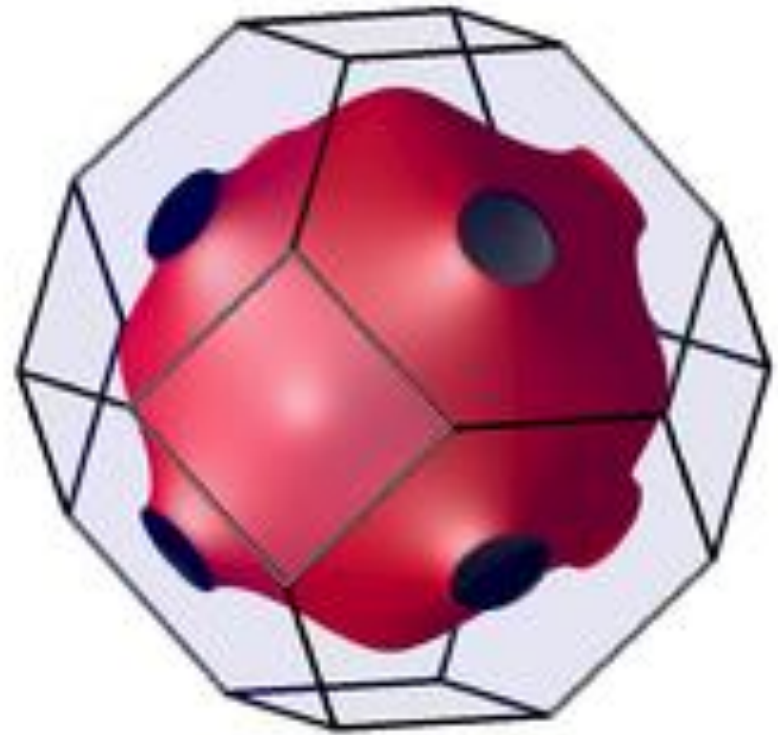


3D case

copper (FCC)



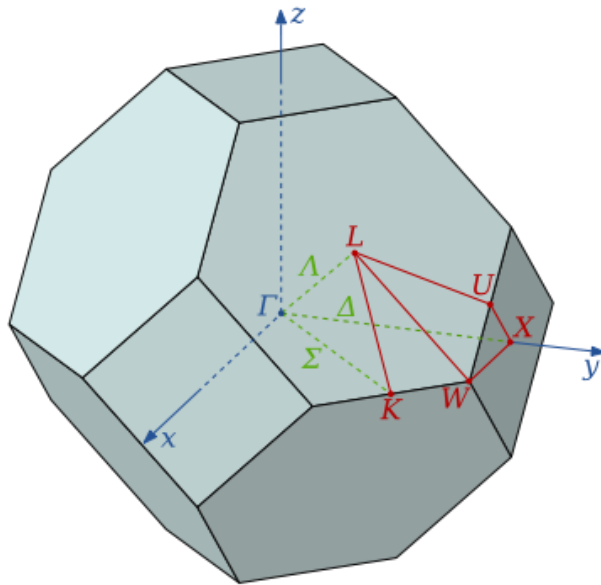
First BZ



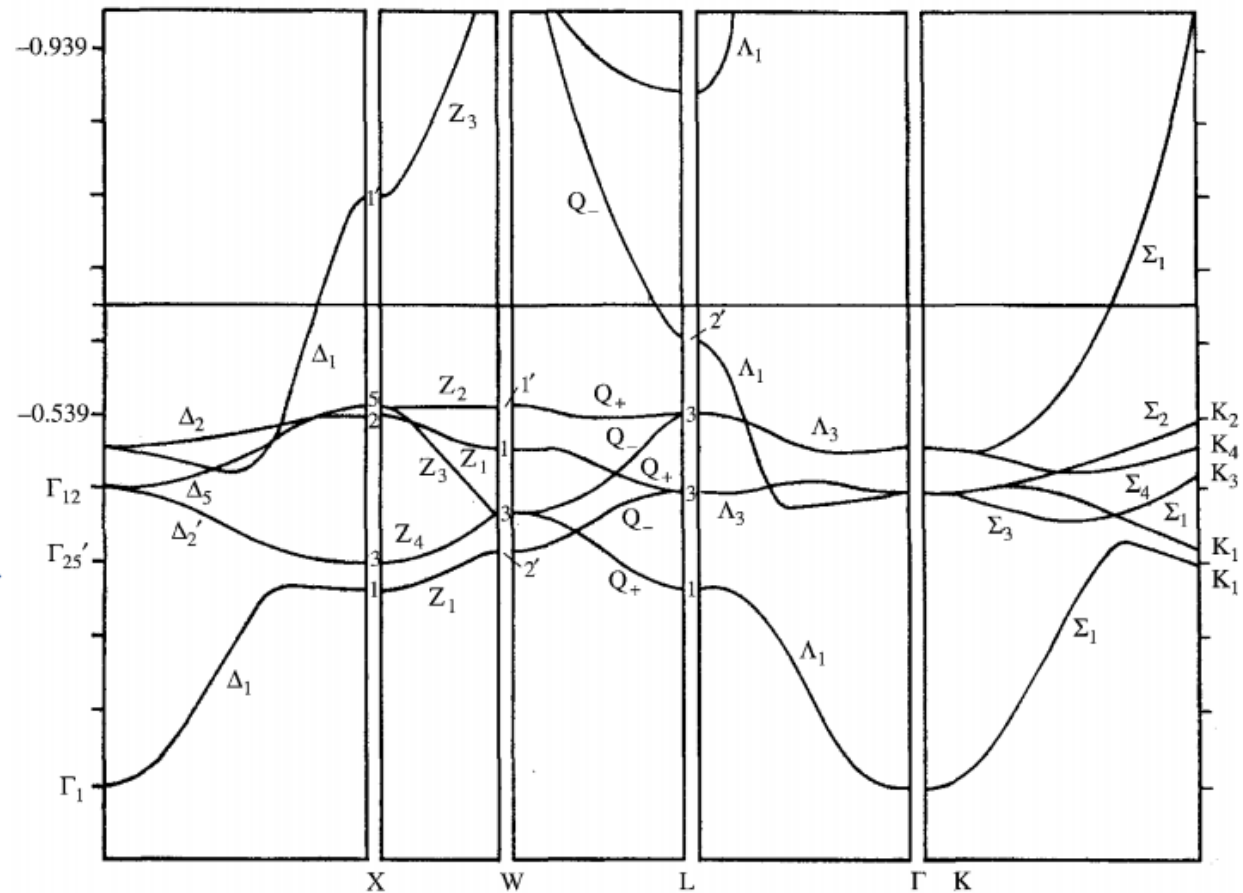
Fermi Surface

3D case

copper (FCC): band structure



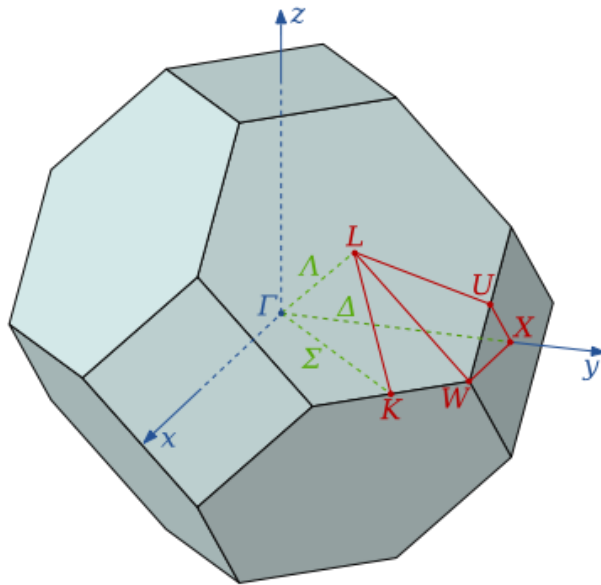
First BZ



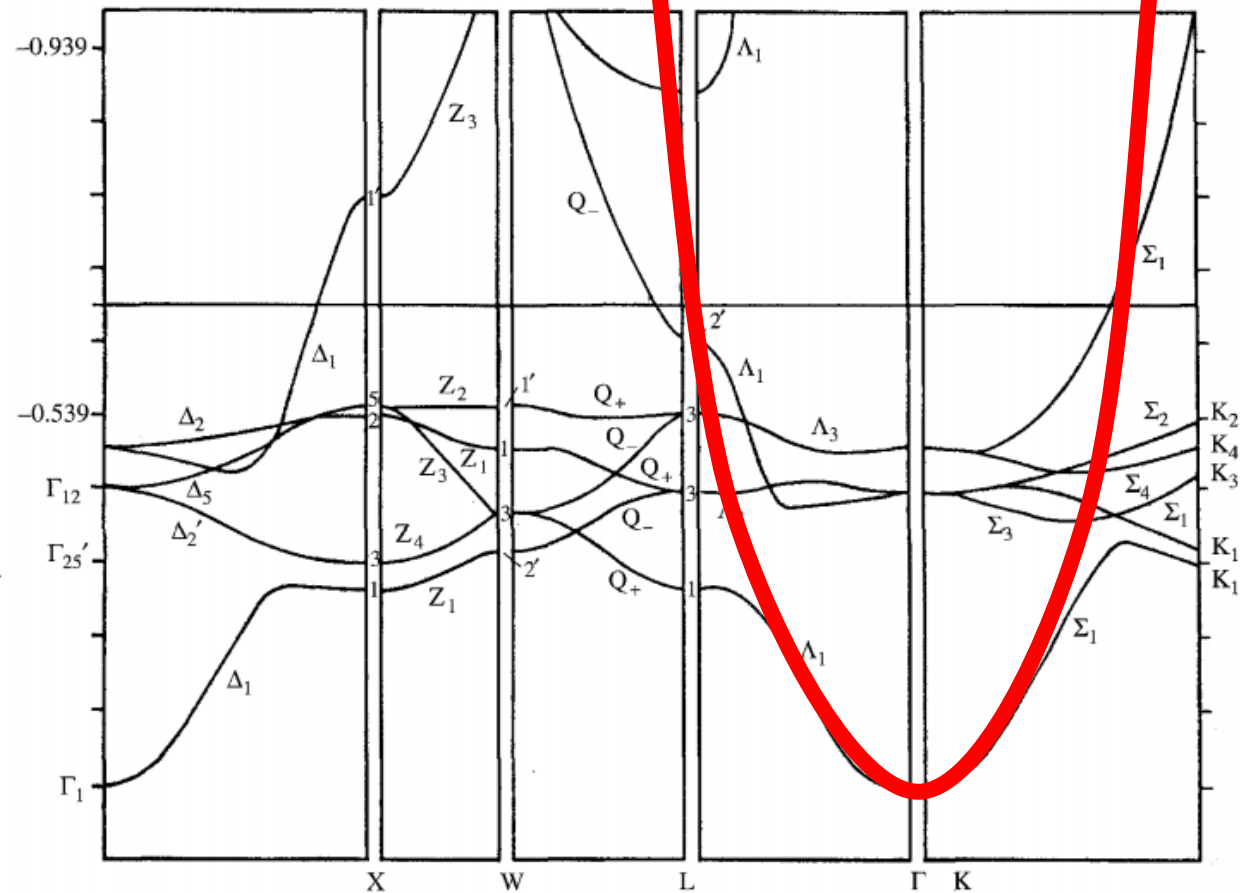
3D case

copper (FCC): band structure

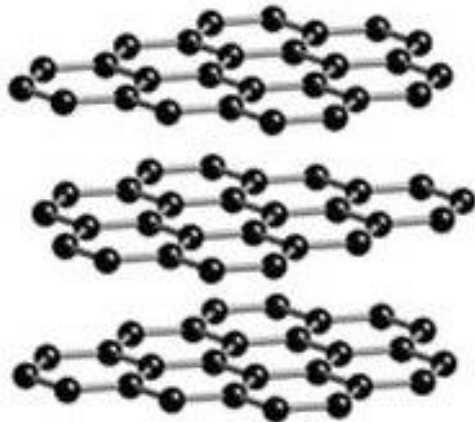
similar to free electrons



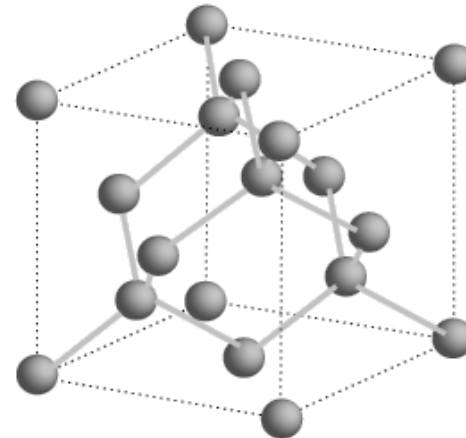
First BZ



Example - Carbon

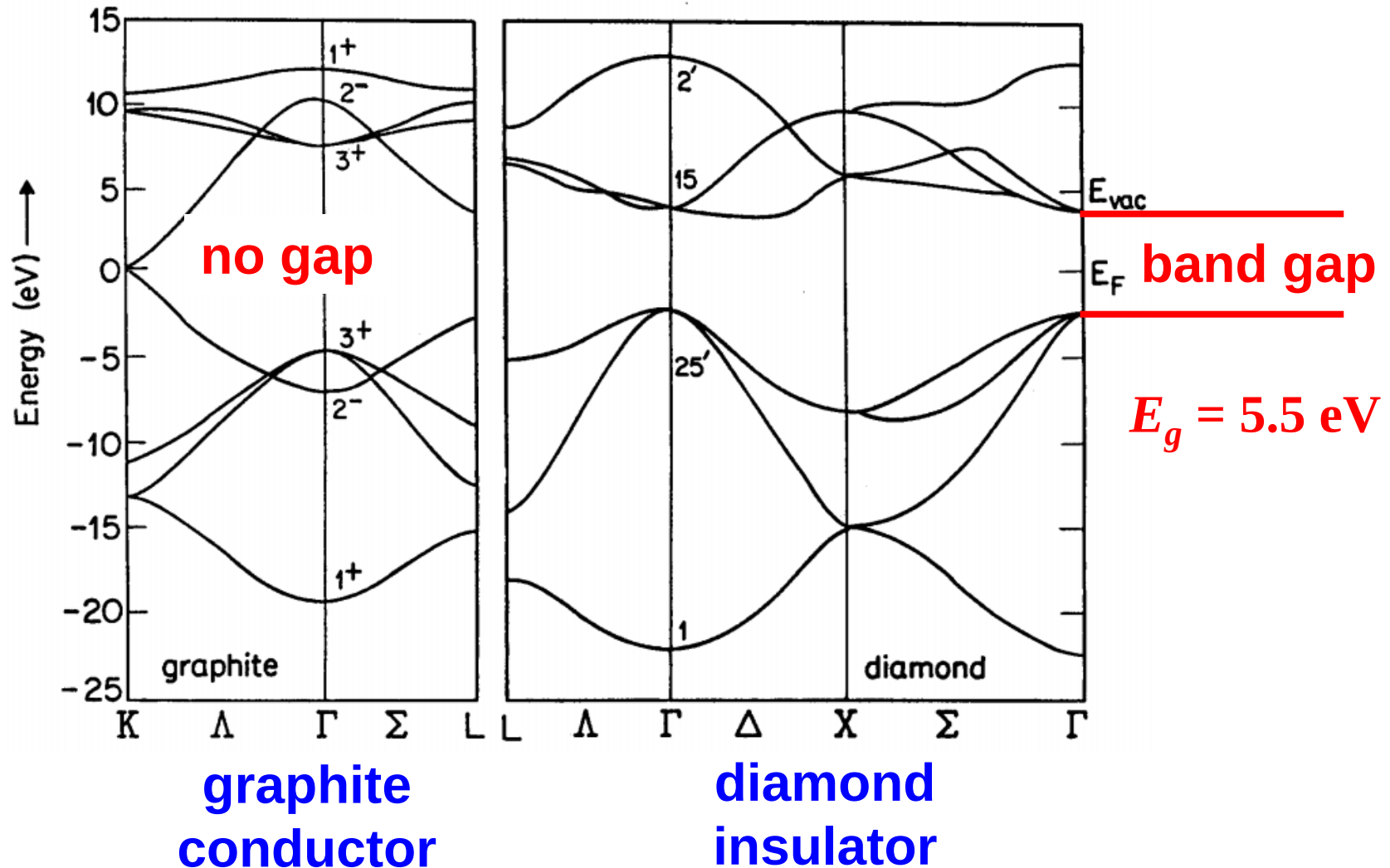


graphite
 $\sigma \sim 10^5 \text{ S/m}$



diamond
 $\sigma \sim 10^{-13} \text{ S/m}$

Example - Carbon



Temperature Dependence of σ

- For metals

- n and m^* have weak dependence on T
- τ has a strong dependence on T
- higher T ---> shorter l and τ ---> smaller μ ---> smaller σ

$$\frac{1}{\rho} = \sigma = ne\mu = \frac{ne^2\tau}{m^*}$$

$$\tau = \frac{l}{v}$$

τ - relaxation time (s)
 l - mean free path (m)

- At $T = 0$ K, perfect metals should have no collision, infinite l and τ ---> infinite σ , zero resistivity
- Resistance is from impurities and thermal vibration.

Temperature Dependence of σ

- Resistivity originates from the collision with imperfect crystals

Matthiessen's rule

$$\rho = \rho_i + \rho_p$$

impurity

phonon
(thermal vibration)

$$\rho_i \sim \frac{1}{l_i} \sim n_i = \text{constant}$$

$$\rho_p \sim \frac{1}{l_p} \sim n_p \sim k_B T$$

(at high T)

$$\rho(T) = \rho_0 [1 + \alpha \Delta T]$$

Temperature Dependence of σ

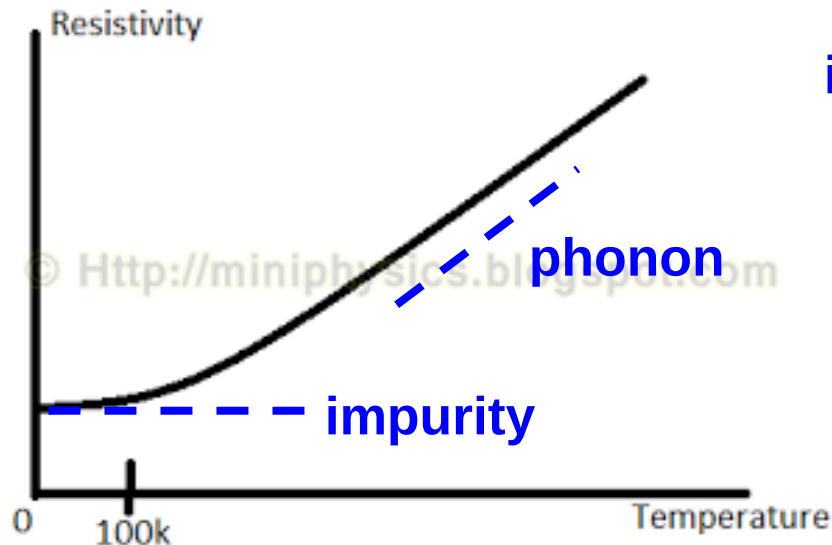
- Resistivity originates from the collision with imperfect crystals

Matthiessen's rule

$$\rho = \rho_i + \rho_p$$

impurity

phonon
(thermal vibration)

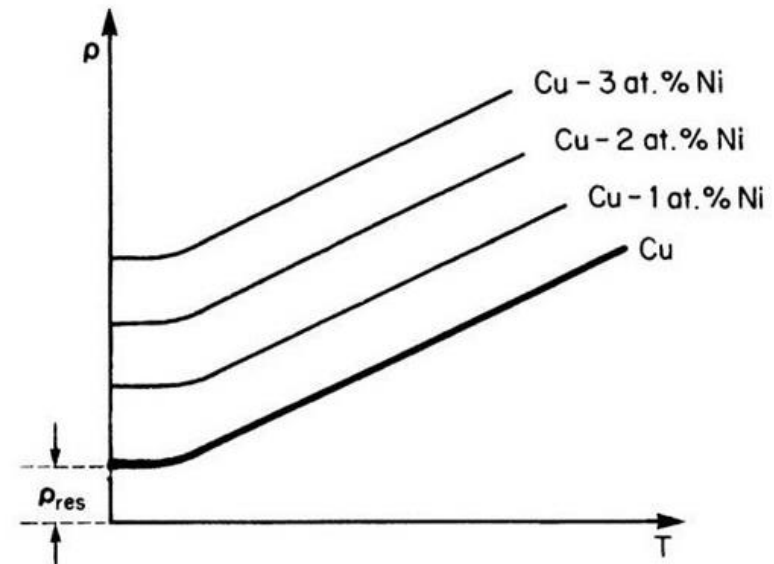
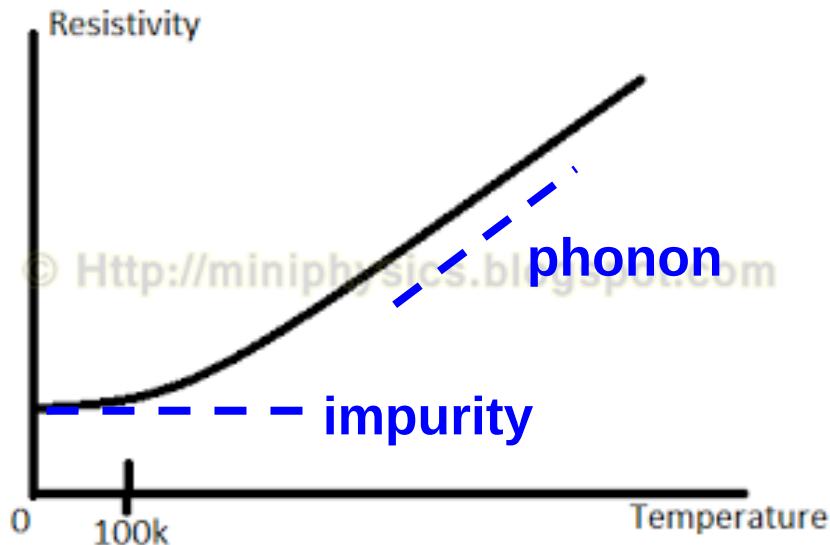


$$\rho(T) = \rho_0[1 + \alpha\Delta T]$$

Temperature Dependence of σ

- Resistivity originates from the collision with imperfect crystals

$$\rho(T) = \rho_0[1 + \alpha\Delta T]$$



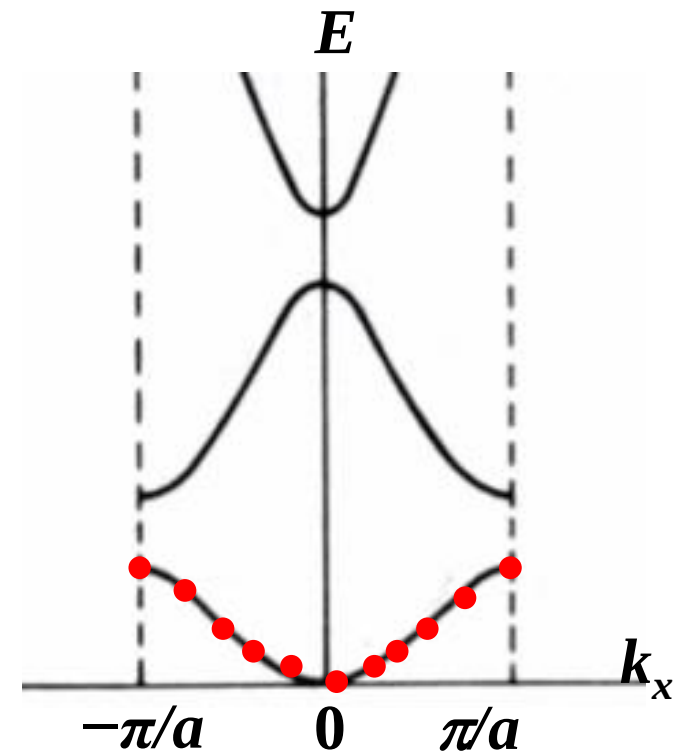
Summary

- Electronic properties of solids depend on

- band structure
- electron density

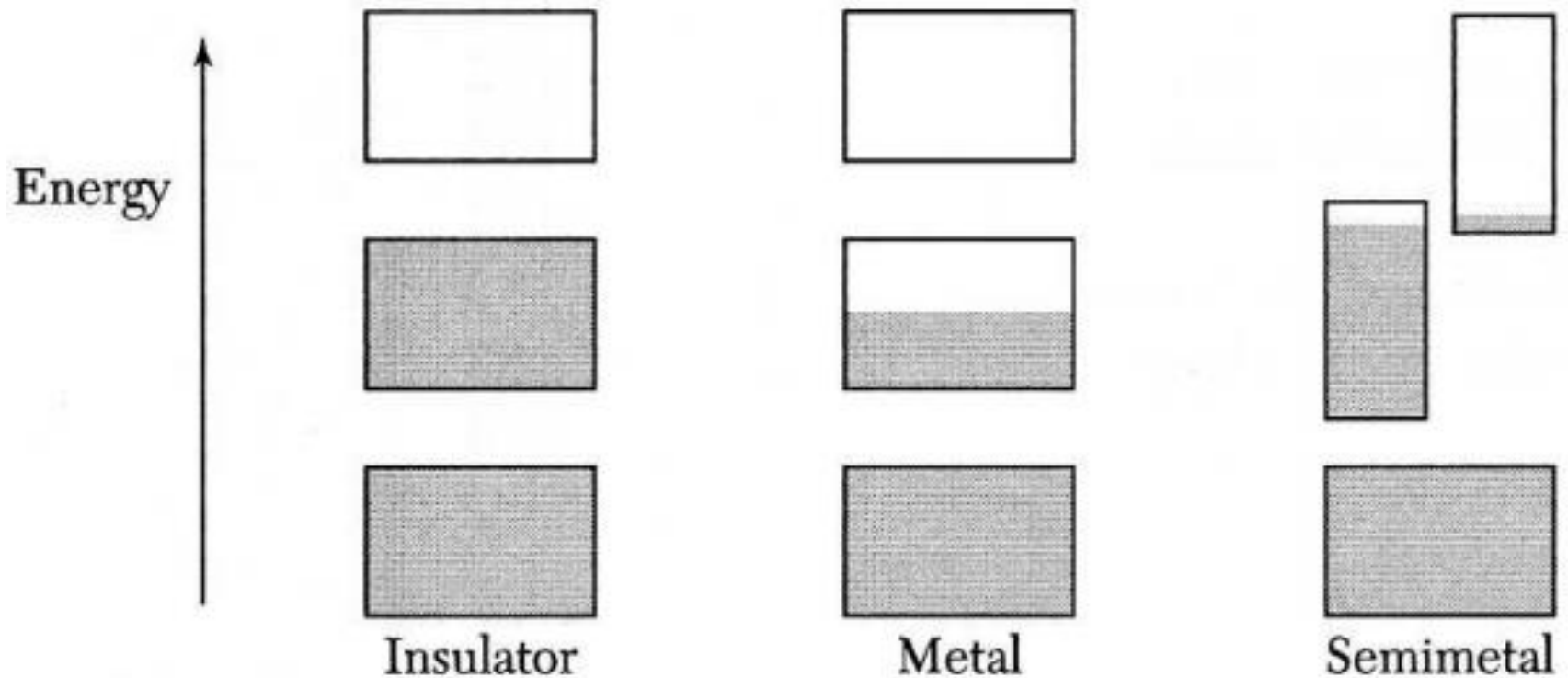
- also depend on

- defects
- temperature
- electric field
- ...



• electron

Summary



Thank you for your attention